

Лабораторная работа №2

Методы нулевого и первого порядка

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Группа : M3233

Описание методов

Не библиотечные методы

get_p_for_newton - метод вычисляет направление шага p в методе Ньютона, решая систему $H \cdot p = -\text{grad}$, и если матрица Гессе вырождена, возвращает $p = -\text{grad}$.

```
def get_p_for_newton(f, x, y, grad, counters):
    H = hess_f(f, x, y)
    counters[0] += 16
    counters[1] += 4
    try:
        p = -np.linalg.solve(H, grad)
    except np.linalg.LinAlgError:
        print("Гессеан вырожден")
        p = -grad

    return p
```

BFGS (Broyden-Fletcher-Goldfarb-Shanno) - квази-Ньютоновский метод для поиска локального минимума функции. Находит точку минимума функции без прямого вычисления гессеана. Алгоритм итеративно формирует и обновляет приближение матрицы, используя информацию о градиентах (1-х производных)

```
def bfgs(f, x, y, B, prev_args, prev_grad, grad, iteration, log_file, counters, c1, a0, q):
    if iteration == 0:
        B = np.eye(2)
        prev_args = np.array([x, y])
        prev_grad = grad
        p = -grad / (np.linalg.norm(grad) + 1e-12)
    else:
        x_prev = prev_args
        s = np.array([x, y]) - x_prev
        y_vec = grad - prev_grad
        Bs = B.dot(s)

        ys = y_vec @ s
        sBs = s @ Bs

        if ys > 1e-12 and sBs > 1e-12:
            B += np.outer(y_vec, y_vec)/ys - np.outer(Bs, Bs)/sBs

        prev_args = np.array([x, y])
        prev_grad = grad

        p = -B.dot(grad)

    direction = np.dot(grad, p)
    x_k, y_k =
```

```

armiyo_gradient_descent(f=f,x=x,y=y,p=p,direction=direction,c1=c1,a0=a0,
    q=q,log_file=log_file,counters=counters)

s2 = np.array([x_k - x, y_k - y])
grad_new = grad_f(f, x_k, y_k)
y_vec2 = grad_new - grad
Bs2 = B.dot(s2)

ys2 = y_vec2 @ s2
sBs2 = s2 @ Bs2

if ys2 > 1e-12 and sBs2 > 1e-12:
    B += np.outer(y_vec2, y_vec2)/ys2 - np.outer(Bs2, Bs2)/sBs2

log_file.write(f"{x_k} {y_k}\n")

return x_k, y_k, -1, -1, B, np.array([x_k, y_k]), grad_new

```

hess_f - метод вычисляет матрицу Гессе (вторых производных) функции f в точке (x, y) методом центральных конечных разностей. Она смещает x и y на малые значения для оценки изменения градиента и формирования матрицы.

```

def hess_f(f, x, y):
    eps = np.finfo(float).eps
    delta_x = np.sqrt(eps) * max(1.00, abs(x))
    delta_y = np.sqrt(eps) * max(1.00, abs(y))

    grad_x_plus = grad_f(f, x + delta_x, y)
    grad_x_minus = grad_f(f, x - delta_x, y)
    grad_y_plus = grad_f(f, x, y + delta_y)
    grad_y_minus = grad_f(f, x, y - delta_y)

    dgrad_dx = (grad_x_plus - grad_x_minus) / (2 * delta_x)
    dgrad_dy = (grad_y_plus - grad_y_minus) / (2 * delta_y)

    H = np.array([
        [dgrad_dx[0], dgrad_dy[0]],
        [dgrad_dx[1], dgrad_dy[1]],
    ])

    return H

```

grad_f - метод приближённо вычисляет градиент функции f в точке (x, y) методом центральных конечных разностей, возвращая вектор, содержащий производные по x и y .

```

def grad_f(f, x, y):
    eps = np.finfo(float).eps
    delta_x = np.sqrt(eps) * max(1.00, abs(x))
    delta_y = np.sqrt(eps) * max(1.00, abs(y))
    df_dx = (f([x + delta_x, y]) - f([x - delta_x, y])) / (2 * delta_x)
    df_dy = (f([x, y + delta_y]) - f([x, y - delta_y])) / (2 * delta_y)

    return np.array([df_dx, df_dy])

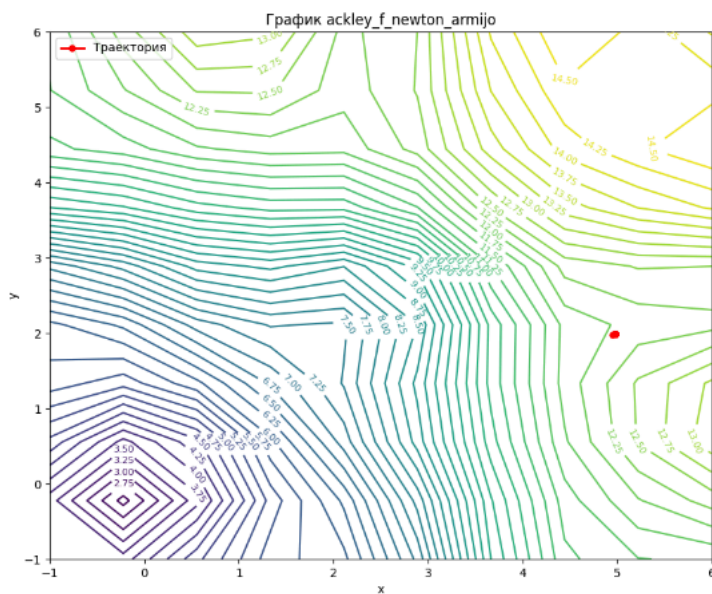
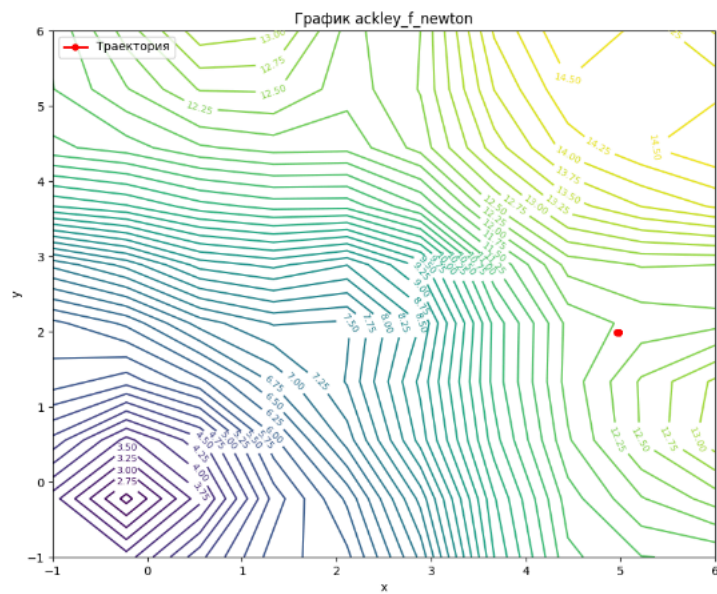
```

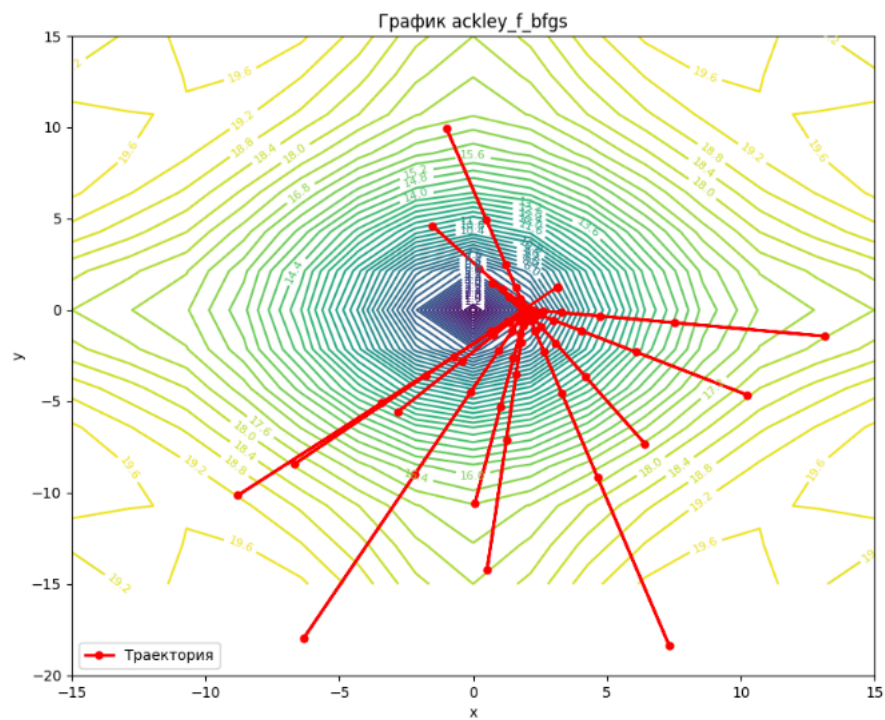
Библиотечные методы

Используются библиотека “*numpy*”, “*SciPy*”

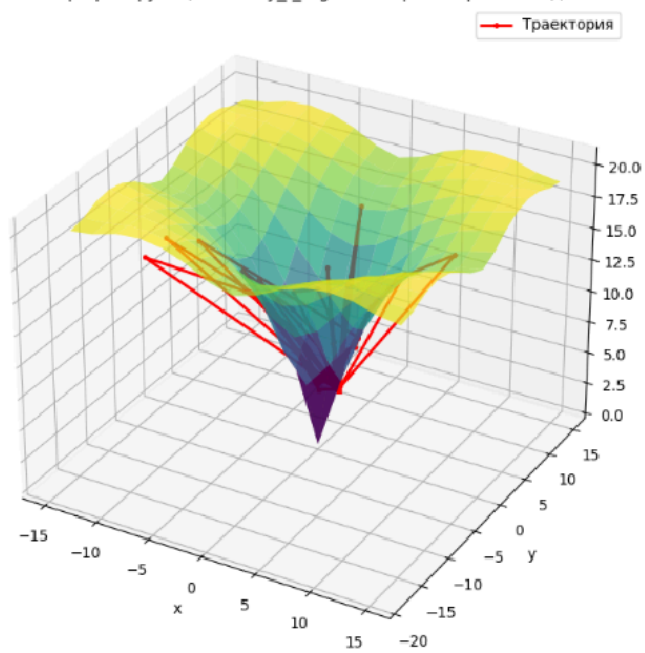
- `np.linalg.solve` решение системы линейных уравнений
- `np.array` создание многомерного массива
- `np.sqrt` вычисление квадратного корня
- `np.outer` вычисление внешнего произведения двух векторов
- `scipy.optimize.minimize` поиск минимума заданной функции
- `np.finfo(float).eps` возвращает машинный эпсилон для типа `float`

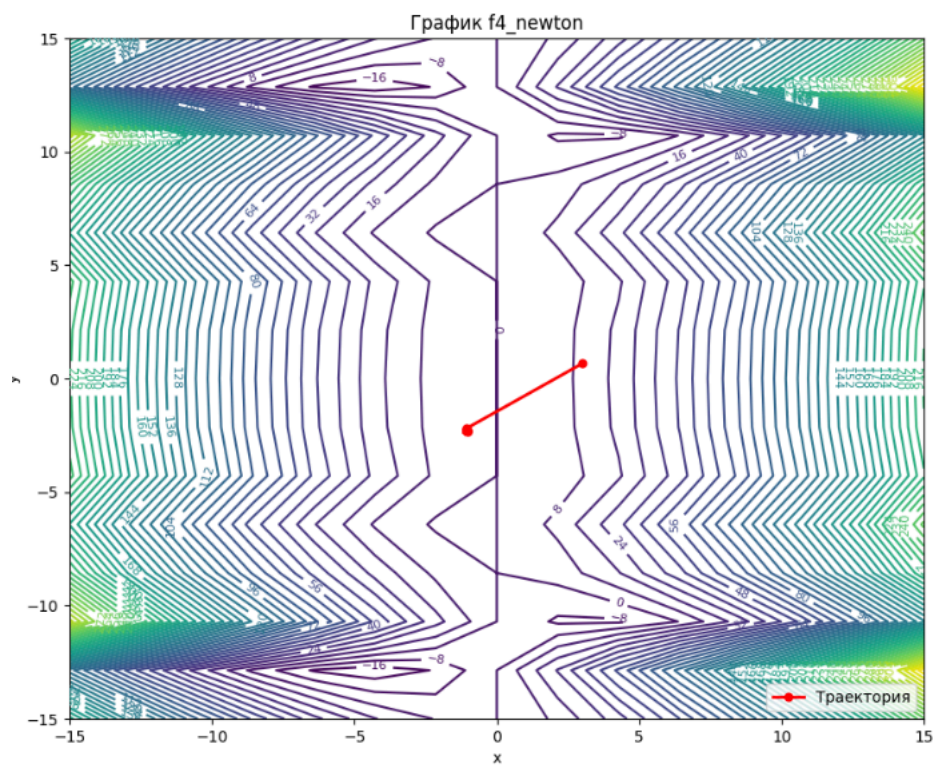
Графики



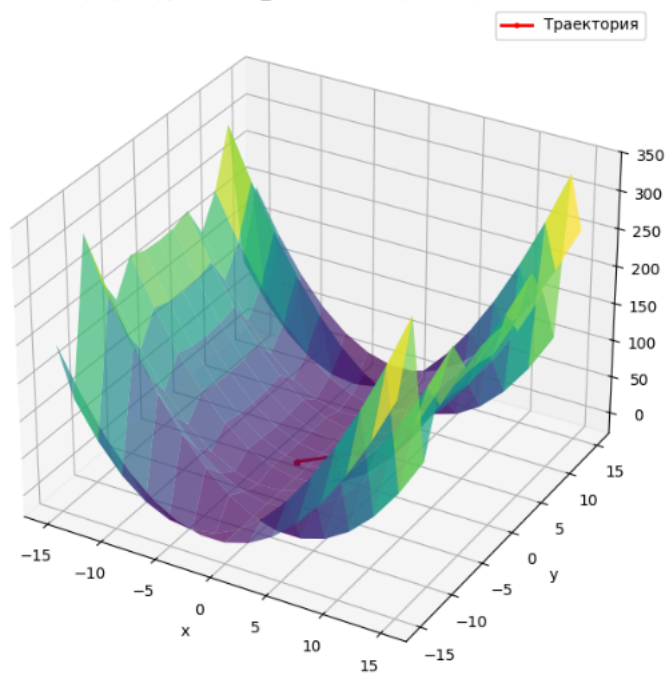


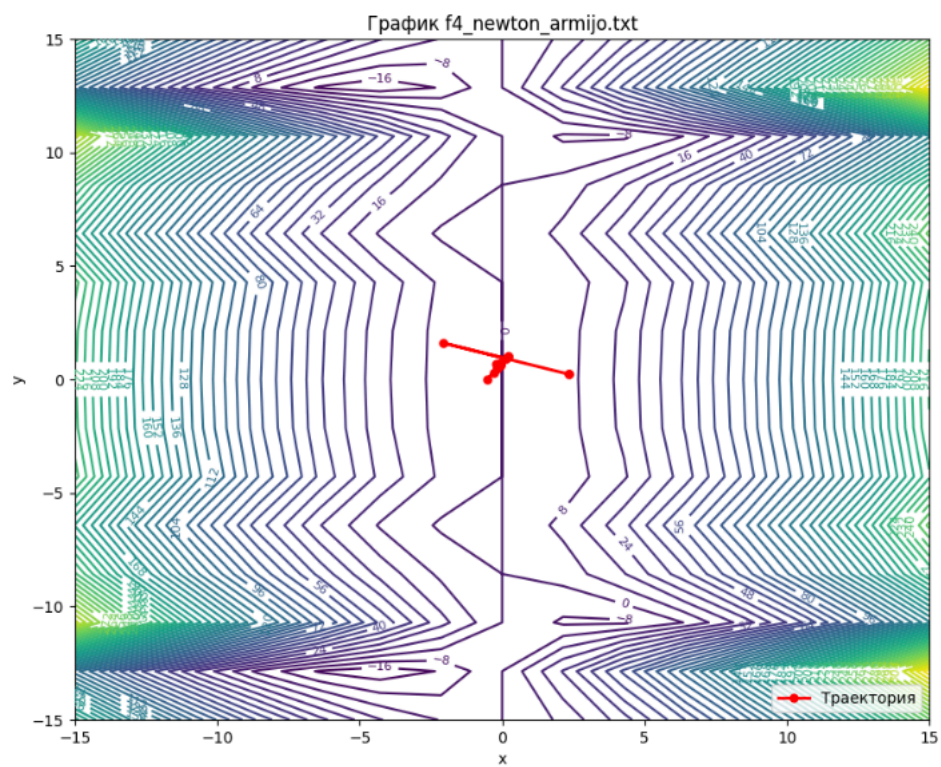
3D график функции ackley_f_bfgs.txt и траектория метода



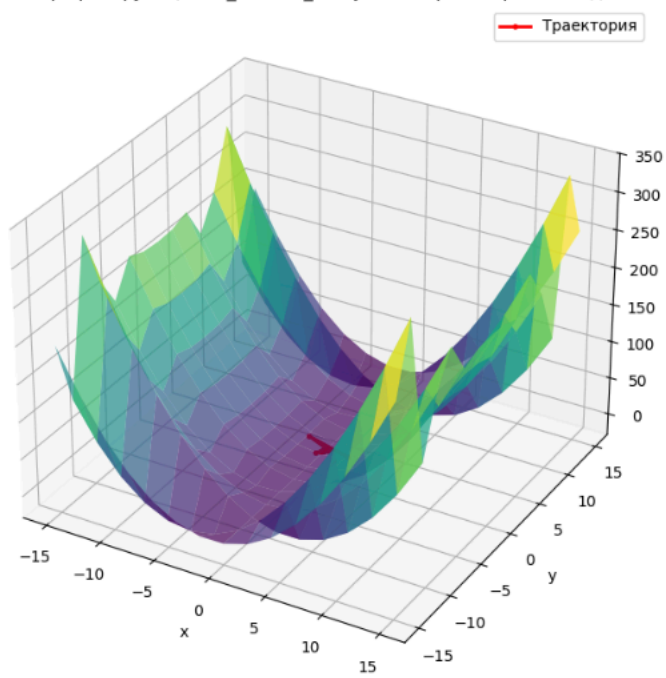


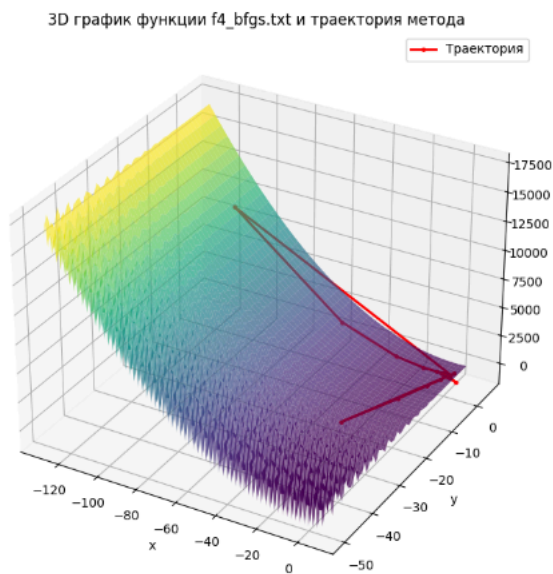
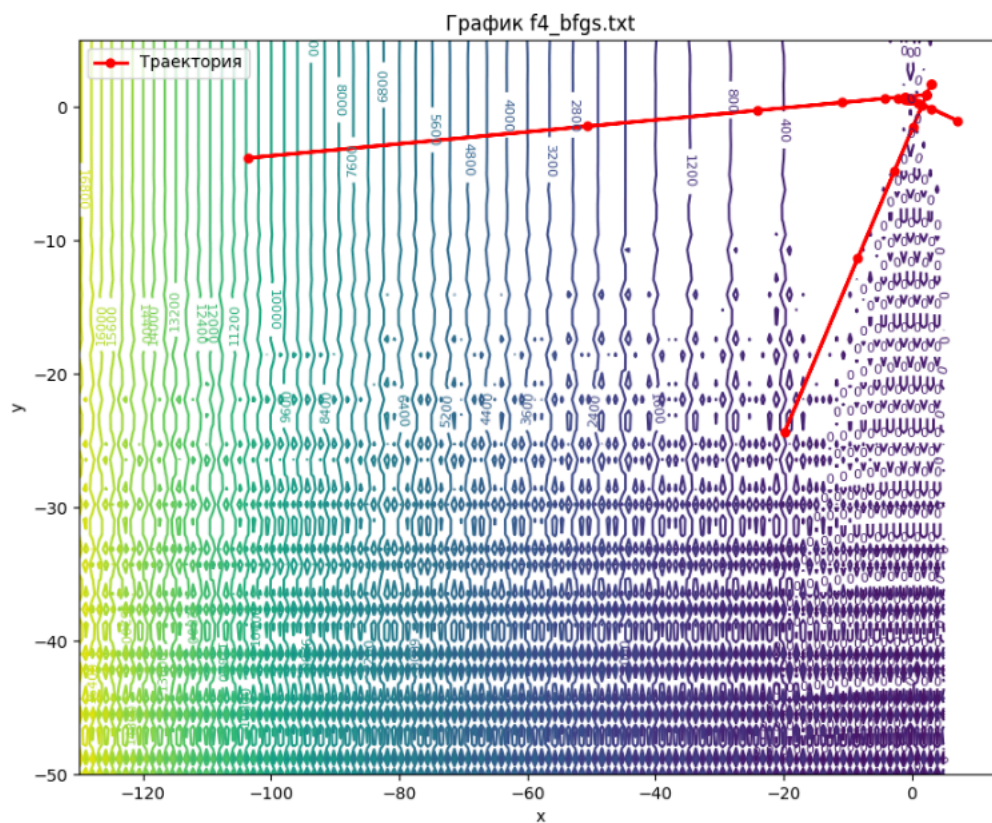
3D график функции f4_newton.txt и траектория метода

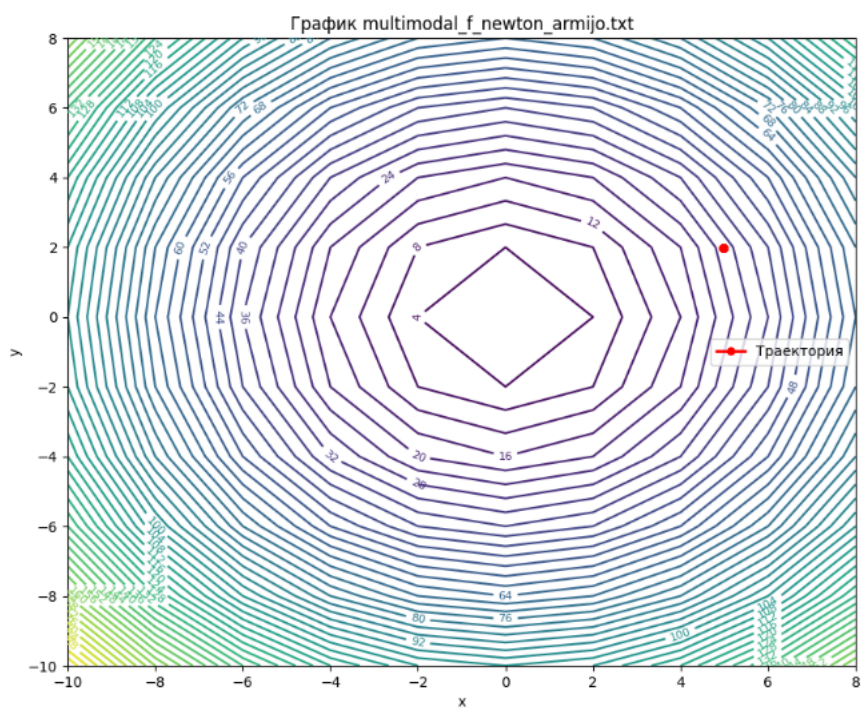
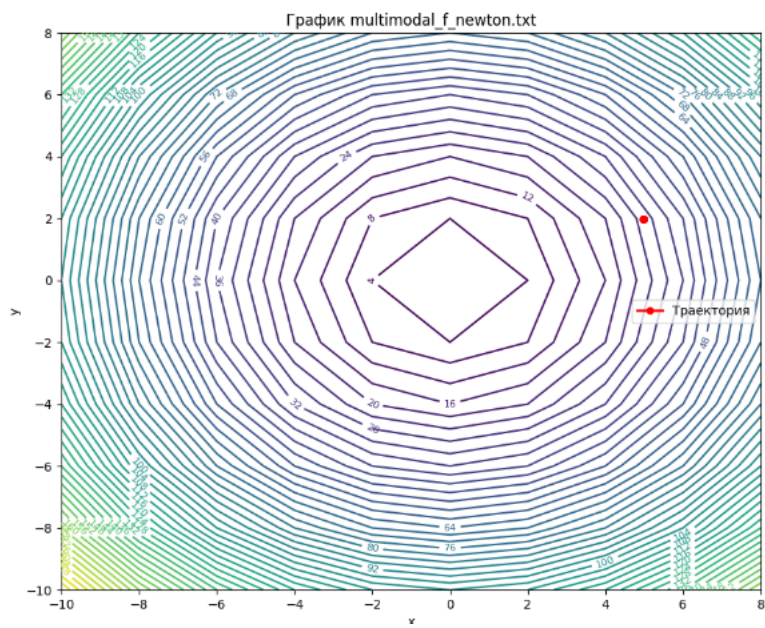


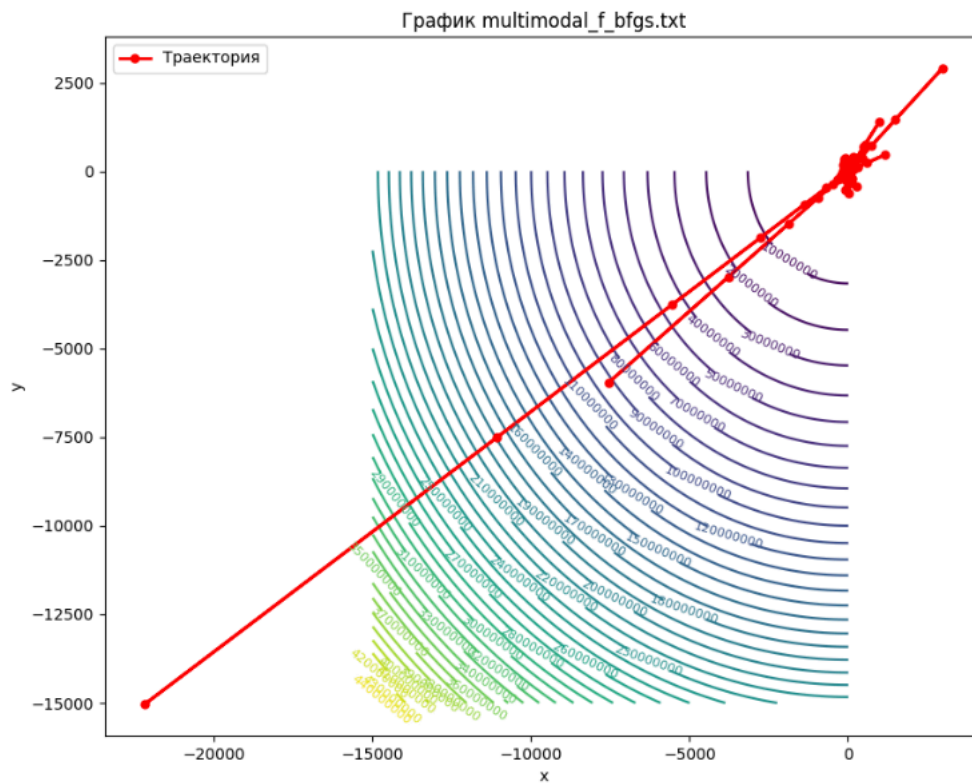


3D график функции f4_newton_armijo.txt и траектория метода

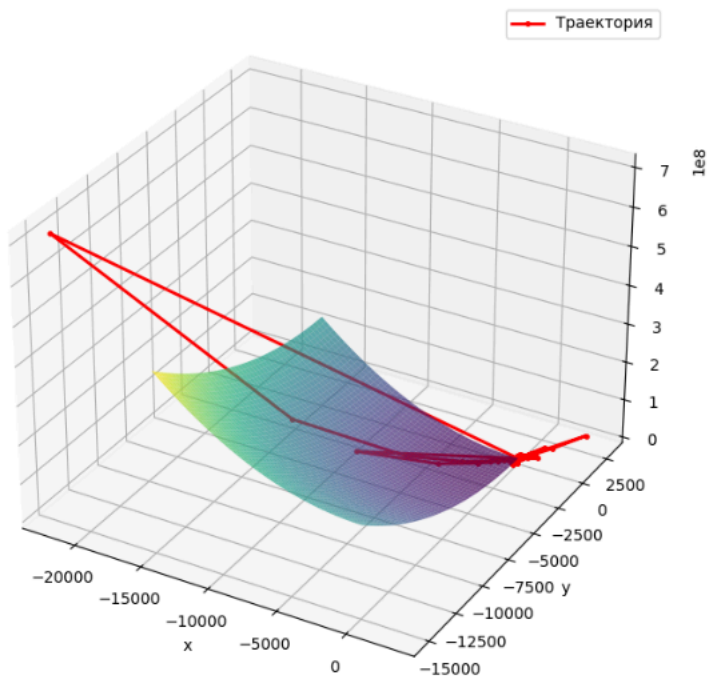




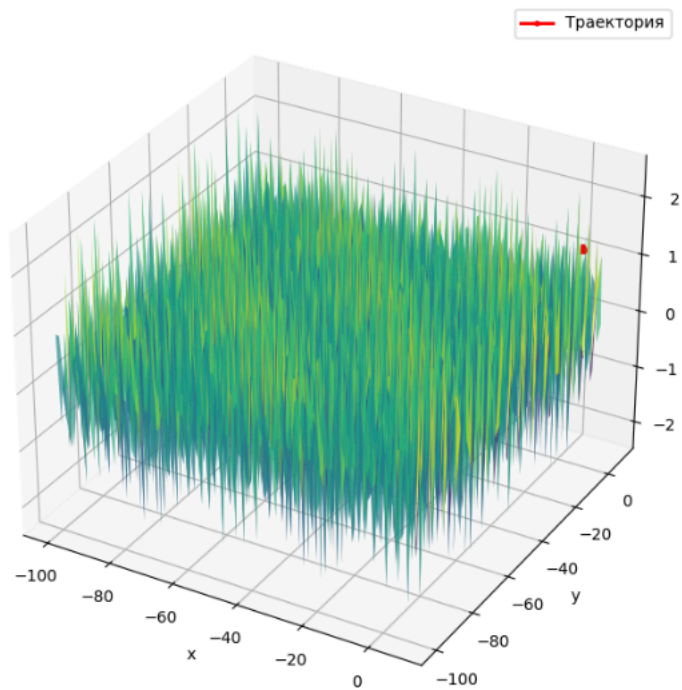




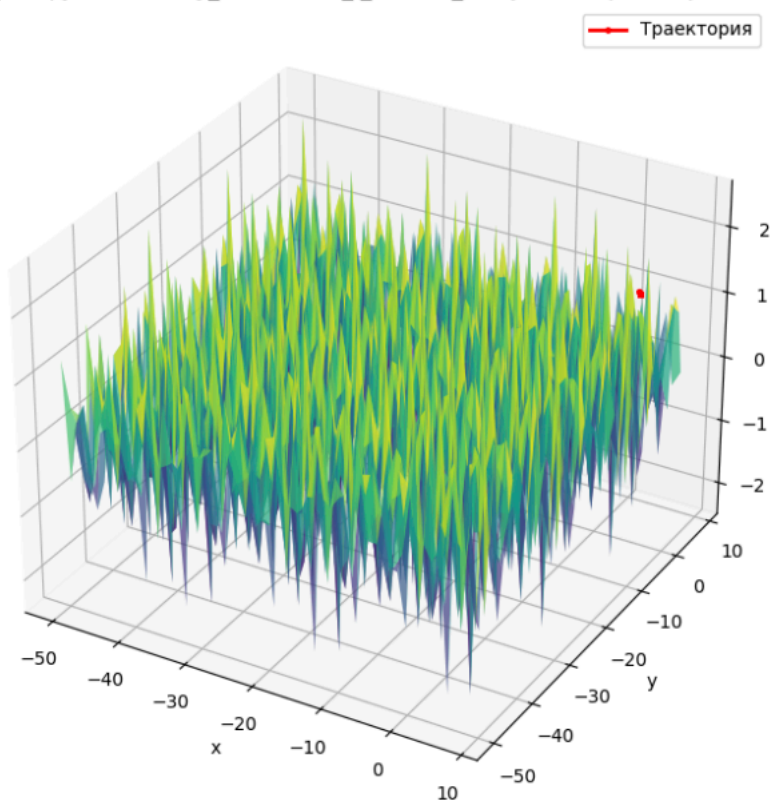
3D график функции multimodal_f_bfgs.txt и траектория метода



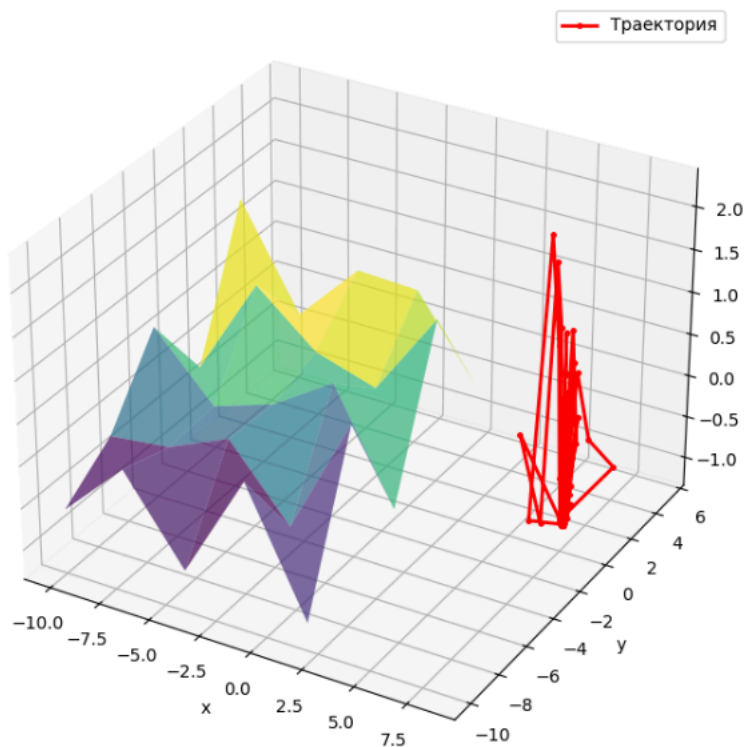
3D график функции noisy_multimodal_f_newton.txt и траектория метода



3D график функции noisy_multimodal_f_newton_armijo.txt и траектория метода



3D график функции noisy_multimodal_f_bfgs.txt и траектория метода



Optuna

Armijo: ackley_f

```
=====
Hyperparameter optimization (Armijo) for: ackley_f = -20 * np.exp(-0.2 * np.sqrt((x**2+y**2)/2)) - np.exp((cos(2*pi*x)+cos(2*pi*y))/2) + 20 + np.e
-----
Best hyperparameters:
  c1      = 0.208850
  a0      = 2.158481
  q       = 0.436254
  iterations= 17046
Best objective value: 4.440892e-16
-----

Trial details:
Trial 0: value = 7.180952e+00, params = {'c1': 0.39409138281053724, 'a0': 1.6502130708896516, 'q': 0.6385995164213529, 'iterations': 21339}
Trial 1: value = 5.381864e+00, params = {'c1': 0.2297698861645607, 'a0': 2.4214391039068737, 'q': 0.4644111608300367, 'iterations': 47139}
Trial 2: value = 1.064503e+01, params = {'c1': 0.2577832473618164, 'a0': 0.6281132446914492, 'q': 0.4980225426455598, 'iterations': 46305}
Trial 3: value = 1.064503e+01, params = {'c1': 0.2850718060050592, 'a0': 0.5827379074978418, 'q': 0.35449585043155113, 'iterations': 43764}
Trial 4: value = 3.996803e-15, params = {'c1': 0.2259787224952095, 'a0': 2.0068756263473575, 'q': 0.4022342857575859, 'iterations': 17231}
Trial 5: value = 1.064503e+01, params = {'c1': 0.21036105677667818, 'a0': 0.5428454319703073, 'q': 0.40942307941206574, 'iterations': 16605}
Trial 6: value = 1.064503e+01, params = {'c1': 0.38333825115616066, 'a0': 1.3568245722048564, 'q': 0.43461511246682477, 'iterations': 35685}
Trial 7: value = 4.440892e-16, params = {'c1': 0.2088504825988655, 'a0': 2.1584812985220347, 'q': 0.4362541407730457, 'iterations': 17046}
Trial 8: value = 5.381864e+00, params = {'c1': 0.20476617356186871, 'a0': 2.2735470880500923, 'q': 0.779051804871062, 'iterations': 41859}
Trial 9: value = 3.996803e-15, params = {'c1': 0.33801507319941504, 'a0': 2.8548403258036017, 'q': 0.7714070684137925, 'iterations': 4894}
Trial 10: value = 5.381864e+00, params = {'c1': 0.11637721193704234, 'a0': 1.3515885251145046, 'q': 0.6066913171705911, 'iterations': 278}
Trial 11: value = 3.574452e+00, params = {'c1': 0.4771140094295597, 'a0': 2.166954833421517, 'q': 0.30068582895796747, 'iterations': 11834}
Trial 12: value = 7.180952e+00, params = {'c1': 0.12799412223529172, 'a0': 1.9119710255075146, 'q': 0.5387570294374189, 'iterations': 29122}
Trial 13: value = 5.381864e+00, params = {'c1': 0.16924787772278482, 'a0': 2.7062179245768947, 'q': 0.37557901677277616, 'iterations': 26675}
Trial 14: value = 7.180952e+00, params = {'c1': 0.33233055965079566, 'a0': 1.6665737493137178, 'q': 0.3308031635218435, 'iterations': 11650}
Trial 15: value = 6.882583e+00, params = {'c1': 0.15892102308585565, 'a0': 1.9490416196984177, 'q': 0.5246716419508644, 'iterations': 19641}
Trial 16: value = 5.381864e+00, params = {'c1': 0.2526057974582783, 'a0': 2.5238831627162126, 'q': 0.6221210776073419, 'iterations': 32903}
Trial 17: value = 1.064503e+01, params = {'c1': 0.3004973853745111, 'a0': 1.2092585195781544, 'q': 0.41595587820932256, 'iterations': 12538}
Trial 18: value = 4.440892e-16, params = {'c1': 0.17556005879604414, 'a0': 2.1283640621410864, 'q': 0.4810386877920888, 'iterations': 23235}
Trial 19: value = 4.884065e+00, params = {'c1': 0.16640091348705008, 'a0': 2.60291529057357, 'q': 0.6902406846105653, 'iterations': 23110}
Trial 20: value = 4.440892e-16, params = {'c1': 0.10392495363064064, 'a0': 2.952761974586309, 'q': 0.4669934845902492, 'iterations': 35663}
Trial 21: value = 4.440892e-16, params = {'c1': 0.1118554118334835, 'a0': 2.9050662534265745, 'q': 0.4782605699698872, 'iterations': 35841}
Trial 22: value = 4.884065e+00, params = {'c1': 0.15024626639278557, 'a0': 2.4070714847068757, 'q': 0.570556220364688, 'iterations': 29288}
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Trial 23: value = 3.996803e-15, params = {'c1': 0.19285731433724118, 'a0': 2.994987039741693, 'q': 0.46207573062943624, 'iterations': 38763}
Trial 24: value = 4.440892e-16, params = {'c1': 0.18061552545254408, 'a0': 2.1790855723811733, 'q': 0.5194184874248329, 'iterations': 24779}
Trial 25: value = 1.064503e+01, params = {'c1': 0.18062385137205156, 'a0': 0.9199862672431485, 'q': 0.4423071338609274, 'iterations': 31862}
Trial 26: value = 5.381864e+00, params = {'c1': 0.13578416959666184, 'a0': 2.6811833026463536, 'q': 0.5736734574855555, 'iterations': 7683}
Trial 27: value = 7.180952e+00, params = {'c1': 0.24497574513278852, 'a0': 1.8625985930912927, 'q': 0.4919497656648345, 'iterations': 18320}
Trial 28: value = 4.440892e-16, params = {'c1': 0.1477656543799707, 'a0': 1.5172278592751298, 'q': 0.3779861238506004, 'iterations': 25873}
Trial 29: value = 7.180952e+00, params = {'c1': 0.18932495517196876, 'a0': 1.7348449351196842, 'q': 0.6703235295755656, 'iterations': 21910}
Trial 30: value = 2.579928e+00, params = {'c1': 0.47288049872480165, 'a0': 2.090080758527432, 'q': 0.5809482890613298, 'iterations': 15032}
Trial 31: value = 5.381864e+00, params = {'c1': 0.10529216266871069, 'a0': 2.348256695877576, 'q': 0.4782520979669711, 'iterations': 36965}
Trial 32: value = 4.440892e-16, params = {'c1': 0.13471407878598018, 'a0': 2.8719993741728738, 'q': 0.447999525988428, 'iterations': 40344}
Trial 33: value = 2.579928e+00, params = {'c1': 0.12029488332397979, 'a0': 2.994885189821775, 'q': 0.4804704195164395, 'iterations': 33363}
Trial 34: value = 2.579928e+00, params = {'c1': 0.22347994433173135, 'a0': 2.7661546758896307, 'q': 0.49748860664907146, 'iterations': 49664}
Trial 35: value = 5.381864e+00, params = {'c1': 0.28498932590964293, 'a0': 2.50422809626998, 'q': 0.3870743855641626, 'iterations': 44431}
Trial 36: value = 4.440892e-16, params = {'c1': 0.10335320927345999, 'a0': 2.2616953739799683, 'q': 0.5128829437887259, 'iterations': 29351}
Trial 37: value = 4.440892e-16, params = {'c1': 0.14783334713678487, 'a0': 2.849737100216055, 'q': 0.4304354943792138, 'iterations': 20184}
Trial 38: value = 5.381864e+00, params = {'c1': 0.20768671928546456, 'a0': 2.5797874962849106, 'q': 0.35084568545909944, 'iterations': 35102}
Trial 39: value = 5.381864e+00, params = {'c1': 0.17755061546193288, 'a0': 2.4193114234525073, 'q': 0.4635815467793738, 'iterations': 15836}
Trial 40: value = 3.574452e+00, params = {'c1': 0.23226392212902716, 'a0': 1.5127738989380553, 'q': 0.5541351635912256, 'iterations': 42856}
Trial 41: value = 2.579928e+00, params = {'c1': 0.1031670309564478, 'a0': 2.1293125146030754, 'q': 0.5171226812718589, 'iterations': 24820}
Trial 42: value = 3.996803e-15, params = {'c1': 0.1345886160011101, 'a0': 2.2609891856622166, 'q': 0.41927595815672636, 'iterations': 22897}
Trial 43: value = 7.180952e+00, params = {'c1': 0.1182676152225357, 'a0': 1.8220188552634036, 'q': 0.5429741785906459, 'iterations': 27595}
Trial 44: value = 4.440892e-16, params = {'c1': 0.1024296848615924, 'a0': 2.0166730138128517, 'q': 0.45975107445681845, 'iterations': 31651}
Trial 45: value = 5.381864e+00, params = {'c1': 0.4190624443466619, 'a0': 2.2415145389359683, 'q': 0.40302173449471557, 'iterations': 23681}
Trial 46: value = 1.064503e+01, params = {'c1': 0.2690942006113686, 'a0': 0.7309843683832118, 'q': 0.5139051215362124, 'iterations': 37651}
Trial 47: value = 4.440892e-16, params = {'c1': 0.1583916221217932, 'a0': 2.77482200899457477, 'q': 0.48876875929406156, 'iterations': 18001}
Trial 48: value = 4.440892e-16, params = {'c1': 0.12553972183489978, 'a0': 2.061777388457392, 'q': 0.5344677315342408, 'iterations': 13805}
Trial 49: value = 5.381864e+00, params = {'c1': 0.19361952970786311, 'a0': 2.631472873797136, 'q': 0.6039086450313272, 'iterations': 34982}
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Armijo: f4

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Hyperparameter optimization (Armijo) for: f4 = x**2 + y**2 * np.sin(3 * x) * np.cos(2 * y)
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Best hyperparameters:
  c1      = 0.248152
  a0      = 2.998292
  q       = 0.479030
  iterations= 31385
Best objective value: -1.493995e+58
-----
Trial details:
Trial 0: value = -7.935310e+00, params = {'c1': 0.1503077845739655, 'a0': 0.9109210389771825, 'q': 0.6534264733076021, 'iterations': 28731}
Trial 1: value = -1.672224e+02, params = {'c1': 0.22983423203008813, 'a0': 1.5826083021725093, 'q': 0.7789577979837974, 'iterations': 8480}
Trial 2: value = 3.445235e+00, params = {'c1': 0.40786153863553865, 'a0': 1.6601246338389517, 'q': 0.7840334654882883, 'iterations': 24314}
Trial 3: value = -7.935310e+00, params = {'c1': 0.42441561073891665, 'a0': 1.6167309409316588, 'q': 0.6184284219204765, 'iterations': 27787}
Trial 4: value = -3.827229e-02, params = {'c1': 0.34307781085700967, 'a0': 0.8217212522570221, 'q': 0.7039473877306492, 'iterations': 48922}
Trial 5: value = -2.654774e+00, params = {'c1': 0.21614532246731538, 'a0': 0.6130740549197453, 'q': 0.5309832550749446, 'iterations': 42271}
Trial 6: value = -3.827229e-02, params = {'c1': 0.46994456524060857, 'a0': 1.6244334327611183, 'q': 0.5629729262326366, 'iterations': 1210}
Trial 7: value = -3.751954e+01, params = {'c1': 0.10084670551939912, 'a0': 2.3128679002978028, 'q': 0.7452525911697749, 'iterations': 37146}
Trial 8: value = -2.654774e+00, params = {'c1': 0.24847926336902348, 'a0': 2.228815772393203, 'q': 0.34780312041708467, 'iterations': 2538}
Trial 9: value = -6.632787e+15, params = {'c1': 0.4947555488514094, 'a0': 2.356002400457082, 'q': 0.3002895595110869, 'iterations': 17972}
Trial 10: value = -4.286845e+21, params = {'c1': 0.3319029373549698, 'a0': 2.7943104642382144, 'q': 0.3209334735310917, 'iterations': 15777}
Trial 11: value = -2.242623e+01, params = {'c1': 0.34475718445586073, 'a0': 2.8759316324760253, 'q': 0.31678974751653, 'iterations': 15663}
Trial 12: value = -3.827229e-02, params = {'c1': 0.48703073304344846, 'a0': 2.9639600331154776, 'q': 0.4040809528551378, 'iterations': 18419}
Trial 13: value = -3.827229e-02, params = {'c1': 0.3049023585822241, 'a0': 2.455842107384619, 'q': 0.43517529636031077, 'iterations': 13806}
Trial 14: value = 3.445235e+00, params = {'c1': 0.3920803455331326, 'a0': 2.597814720325998, 'q': 0.45217240244035395, 'iterations': 19757}
Trial 15: value = -2.592684e+03, params = {'c1': 0.49679884346888176, 'a0': 2.0646954313305277, 'q': 0.30501997854310214, 'iterations': 11416}
Trial 16: value = -5.377940e+53, params = {'c1': 0.2826324282981659, 'a0': 2.693687890756241, 'q': 0.3790110836824316, 'iterations': 7160}
Trial 17: value = -1.935122e+02, params = {'c1': 0.28724713806263086, 'a0': 2.7022194771081014, 'q': 0.3836262555896273, 'iterations': 5689}
Trial 18: value = -2.654774e+00, params = {'c1': 0.1699683816760553, 'a0': 1.2514312227334103, 'q': 0.4877400026752522, 'iterations': 8331}
Trial 19: value = -2.025425e+01, params = {'c1': 0.34943930426303477, 'a0': 1.9549487024063956, 'q': 0.3722689430268835, 'iterations': 34354}
Trial 20: value = -1.008242e+01, params = {'c1': 0.2808714269074295, 'a0': 2.7012333372951427, 'q': 0.4892066635800582, 'iterations': 23027}
Trial 21: value = -3.827229e-02, params = {'c1': 0.3797097737326135, 'a0': 2.442689035647302, 'q': 0.33446031463598547, 'iterations': 12430}
Trial 22: value = -1.682158e+53, params = {'c1': 0.44680880282767693, 'a0': 2.995049667844814, 'q': 0.30197643561786697, 'iterations': 18481}
```

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Trial 23: value = -3.827229e-02, params = {'c1': 0.4468887076157453, 'a0': 2.9882627488549045, 'q': 0.4085718483463881, 'iterations': 5931}
Trial 24: value = -5.154656e+03, params = {'c1': 0.3144424808405414, 'a0': 2.7257112399266736, 'q': 0.3560185099067292, 'iterations': 20331}
Trial 25: value = -7.130154e+04, params = {'c1': 0.36704516390004505, 'a0': 2.8386852697507585, 'q': 0.443455915393627, 'iterations': 10735}
Trial 26: value = -3.751954e+01, params = {'c1': 0.26941154611659746, 'a0': 2.0998468427146832, 'q': 0.35236094569143095, 'iterations': 14935}
Trial 27: value = -7.935310e+00, params = {'c1': 0.44368405239878805, 'a0': 2.5735303881644276, 'q': 0.39573051825808103, 'iterations': 26627}
Trial 28: value = -2.654774e+00, params = {'c1': 0.19087988199015854, 'a0': 2.814995827173653, 'q': 0.3000962480902833, 'iterations': 22127}
Trial 29: value = -1.834425e+07, params = {'c1': 0.3104746634525742, 'a0': 1.869005320914086, 'q': 0.6158997459422302, 'iterations': 4400}
Trial 30: value = -1.493995e+58, params = {'c1': 0.24815245807215885, 'a0': 2.99829207610882, 'q': 0.4790302116349868, 'iterations': 31385}
Trial 31: value = -2.654774e+00, params = {'c1': 0.2474116802338997, 'a0': 2.9957515366810274, 'q': 0.4806063903150855, 'iterations': 31686}
Trial 32: value = -3.827229e-02, params = {'c1': 0.20392006170962698, 'a0': 2.541208124567843, 'q': 0.5206582858877732, 'iterations': 28850}
Trial 33: value = -3.970066e+01, params = {'c1': 0.2474648789251913, 'a0': 2.77709439160772, 'q': 0.33566613801116585, 'iterations': 36558}
Trial 34: value = -1.008242e+01, params = {'c1': 0.13759161371793083, 'a0': 1.3963476936053818, 'q': 0.4309629086343618, 'iterations': 8771}
Trial 35: value = -3.533393e+02, params = {'c1': 0.41672069277498036, 'a0': 2.6115210568685208, 'q': 0.5705208686219037, 'iterations': 29612}
Trial 36: value = -8.501302e+03, params = {'c1': 0.22678030870351693, 'a0': 2.8533237898749713, 'q': 0.374412663737135, 'iterations': 25052}
Trial 37: value = 3.445235e+00, params = {'c1': 0.32968008156398626, 'a0': 1.0312598976718819, 'q': 0.3263167550035931, 'iterations': 17210}
Trial 38: value = -2.654774e+00, params = {'c1': 0.26803898741325693, 'a0': 2.2497888266356614, 'q': 0.46866004440539066, 'iterations': 31986}
Trial 39: value = 1.753076e+01, params = {'c1': 0.39587089726020963, 'a0': 2.464398722692345, 'q': 0.5238271209487992, 'iterations': 48404}
Trial 40: value = -2.253664e+47, params = {'c1': 0.36429637516890684, 'a0': 2.9146280347032647, 'q': 0.4165061771846898, 'iterations': 42654}
Trial 41: value = -3.827229e-02, params = {'c1': 0.35841131576564045, 'a0': 2.9047306359467533, 'q': 0.42305484567278695, 'iterations': 42576}
Trial 42: value = 3.445235e+00, params = {'c1': 0.3267760902906674, 'a0': 2.707248406673457, 'q': 0.3670668377671892, 'iterations': 45376}
Trial 43: value = -3.827229e-02, params = {'c1': 0.4565458954600972, 'a0': 2.998470096586513, 'q': 0.39530646795925833, 'iterations': 39047}
Trial 44: value = -3.641295e+00, params = {'c1': 0.293565316464286876, 'a0': 2.871719067062565, 'q': 0.33011725067320336, 'iterations': 42614}
Trial 45: value = -2.643125e+03, params = {'c1': 0.4241421323467628, 'a0': 2.6391071557564882, 'q': 0.45833254244553373, 'iterations': 734}
Trial 46: value = -3.827229e-02, params = {'c1': 0.265691527057962, 'a0': 2.7790033747533673, 'q': 0.416896512390207, 'iterations': 39423}
Trial 47: value = 3.445235e+00, params = {'c1': 0.3287276067322663, 'a0': 2.3922783451323077, 'q': 0.3536548154594946, 'iterations': 21629}
Trial 48: value = 3.445235e+00, params = {'c1': 0.4012707493377327, 'a0': 2.9004476771273486, 'q': 0.6858407485681233, 'iterations': 16453}
Trial 49: value = 3.445235e+00, params = {'c1': 0.380649308643995, 'a0': 2.1453858947750586, 'q': 0.38595729479704743, 'iterations': 13917}
=====
```

Newton_armijo: ackley_f

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=====
Hyperparameter optimization (newton_armijo) for: ackley_f = -20 * np.exp(-0.2 * np.sqrt((x^2+y^2)/2)) - np.exp((cos(2*pi*x)+cos(2*pi*y))/2) + 20 + np.e
-----
Best hyperparameters:
  newton_h = 1.773109
  c1       = 0.489934
  a0       = 1.903023
  q        = 0.632838
  iterations= 35432
Best objective value: 1.064503e+01
-----
Trial details:
Trial 0: value = 1.064503e+01, params = {'newton_h': 0.9323562193927101, 'c1': 0.3455002791604671, 'a0': 2.8092185846189626, 'q': 0.6688105836266994, 'iterations': 4701}
Trial 1: value = 1.064503e+01, params = {'newton_h': 1.7731093243662124, 'c1': 0.4899337351138363, 'a0': 1.903022891325775, 'q': 0.6328382304845528, 'iterations': 35432}
Trial 2: value = 1.064503e+01, params = {'newton_h': 1.9002919365741997, 'c1': 0.17203964680679484, 'a0': 1.1599110348302784, 'q': 0.49903630651779335, 'iterations': 465}
Trial 3: value = 1.064503e+01, params = {'newton_h': 0.12350072575205646, 'c1': 0.3800441553716544, 'a0': 2.4081390842274337, 'q': 0.3124255734879211, 'iterations': 379}
Trial 4: value = 1.064503e+01, params = {'newton_h': 2.5122209568963565, 'c1': 0.22051591651217717, 'a0': 0.748535320564417, 'q': 0.541379633391707, 'iterations': 10306}
Trial 5: value = 1.064503e+01, params = {'newton_h': 1.3657550474572302, 'c1': 0.3517109032150541, 'a0': 1.2282360252896425, 'q': 0.6132094972715142, 'iterations': 2124}
Trial 6: value = 1.064503e+01, params = {'newton_h': 2.598698758820789, 'c1': 0.36580673993148694, 'a0': 2.54389540300083396, 'q': 0.44018180475197793, 'iterations': 409}
Trial 7: value = 1.064503e+01, params = {'newton_h': 1.65408859954717519, 'c1': 0.18185420880358333, 'a0': 2.0952247310176193, 'q': 0.6740403580689173, 'iterations': 6695}
Trial 8: value = 1.064503e+01, params = {'newton_h': 1.9119200621164887, 'c1': 0.10167950346380601, 'a0': 1.54417493569523303, 'q': 0.3455544207362879, 'iterations': 4804}
Trial 9: value = 1.064503e+01, params = {'newton_h': 1.5913064597803537, 'c1': 0.2156255119687289, 'a0': 1.8709280325580633, 'q': 0.6210216438560757, 'iterations': 1165}
Trial 10: value = 1.064503e+01, params = {'newton_h': 0.800987059310027, 'c1': 0.4977167011525063, 'a0': 2.9751227266558007, 'q': 0.7863818318372106, 'iterations': 29917}
Trial 11: value = 1.064503e+01, params = {'newton_h': 0.2351219673616125, 'c1': 0.4862043879615397, 'a0': 2.2623866033852766, 'q': 0.30300579232341807, 'iterations': 322}
Trial 12: value = 1.064503e+01, params = {'newton_h': 0.24950254778113612, 'c1': 0.42207069635110916, 'a0': 2.425364041442835, 'q': 0.422667047935653, 'iterations': 36475}
Trial 13: value = 1.064503e+01, params = {'newton_h': 1.045956829010082, 'c1': 0.4274354997986493, 'a0': 1.8023772583213575, 'q': 0.7880683524350013, 'iterations': 22912}
Trial 14: value = 1.064503e+01, params = {'newton_h': 2.1819907808528978, 'c1': 0.4258815882079632, 'a0': 1.4740025545611393, 'q': 0.40931825825173074, 'iterations': 389}
Trial 15: value = 1.064503e+01, params = {'newton_h': 2.965424183370307, 'c1': 0.2740543977315506, 'a0': 2.102663992240032, 'q': 0.5133826554908715, 'iterations': 30215}
Trial 16: value = 1.064503e+01, params = {'newton_h': 0.6621026658635347, 'c1': 0.46909612778306375, 'a0': 2.6520186899544087, 'q': 0.7151462833298994, 'iterations': 189}
Trial 17: value = 1.064503e+01, params = {'newton_h': 1.297545221858827, 'c1': 0.30741404865187327, 'a0': 2.013886353886615, 'q': 0.5947758604810016, 'iterations': 33732}
Trial 18: value = 1.064503e+01, params = {'newton_h': 0.41582986191248095, 'c1': 0.39048419679337687, 'a0': 2.350545357381561, 'q': 0.46961622813603165, 'iterations': 415}
Trial 19: value = 1.064503e+01, params = {'newton_h': 2.188685522198834, 'c1': 0.45569941865683394, 'a0': 0.7242080057751727, 'q': 0.35828111155251563, 'iterations': 2615}
Trial 20: value = 1.064503e+01, params = {'newton_h': 1.143422565893563, 'c1': 0.3070493518426729, 'a0': 1.5846901186503572, 'q': 0.571175398991955, 'iterations': 1834}
Trial 21: value = 1.064503e+01, params = {'newton_h': 0.6673584023447299, 'c1': 0.461431729186598, 'a0': 2.689114894280421, 'q': 0.712123635220894, 'iterations': 13835}
```

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frial 23: value = 1.064503e+01, params = {'newton_h': 0.15040107095537053, 'c1': 0.46786494151518043, 'a0': 2.918115437125992, 'q': 0.7235544617601064, 'iterations': 26523}
frial 24: value = 1.064503e+01, params = {'newton_h': 0.4981569542000954, 'c1': 0.43587156131098603, 'a0': 2.2751099276399036, 'q': 0.6498921961910933, 'iterations': 18887}
frial 25: value = 1.064503e+01, params = {'newton_h': 0.7318746042975277, 'c1': 0.49939075061884064, 'a0': 2.533114102090002, 'q': 0.7657728103102807, 'iterations': 36904}
frial 26: value = 1.064503e+01, params = {'newton_h': 0.11496261349490522, 'c1': 0.3910930928602663, 'a0': 2.167273351128151, 'q': 0.7447630428882621, 'iterations': 44302}
frial 27: value = 1.064503e+01, params = {'newton_h': 1.780475068823938, 'c1': 0.44891863029688805, 'a0': 1.853587846775231, 'q': 0.6899689696483937, 'iterations': 27738}
frial 28: value = 1.064503e+01, params = {'newton_h': 0.3502322933881013, 'c1': 0.4034091058759722, 'a0': 2.4848775927409637, 'q': 0.6454610753123822, 'iterations': 33615}
frial 29: value = 1.064503e+01, params = {'newton_h': 0.8953437654767945, 'c1': 0.3329948329309679, 'a0': 2.7620229628581807, 'q': 0.5600612158909368, 'iterations': 49932}
frial 30: value = 1.064503e+01, params = {'newton_h': 1.4106781790194327, 'c1': 0.4745588668841553, 'a0': 1.9701899219712482, 'q': 0.6462083147476955, 'iterations': 43079}
frial 31: value = 1.064503e+01, params = {'newton_h': 1.1287803015453768, 'c1': 0.31117569472899904, 'a0': 1.5895553838401988, 'q': 0.5750381946470845, 'iterations': 343}
frial 32: value = 1.064503e+01, params = {'newton_h': 1.20894349584843985, 'c1': 0.2734180785072254, 'a0': 1.2731101444553172, 'q': 0.5268536535756918, 'iterations': 6425}
frial 33: value = 1.064503e+01, params = {'newton_h': 0.9404411438187767, 'c1': 0.369948356324272, 'a0': 1.072800029387217, 'q': 0.4921311623337957, 'iterations': 1183}
frial 34: value = 1.064503e+01, params = {'newton_h': 1.4417266968021853, 'c1': 0.33826410542039353, 'a0': 1.6607053766406172, 'q': 0.5898811244233414, 'iterations': 17859}
frial 35: value = 1.064503e+01, params = {'newton_h': 0.5511592056744344, 'c1': 0.2520984345872512, 'a0': 1.3796925734974501, 'q': 0.6211071763636118, 'iterations': 6664}
frial 36: value = 1.064503e+01, params = {'newton_h': 1.9809232386189248, 'c1': 0.3273585600714013, 'a0': 0.8591323167595838, 'q': 0.5537808925112002, 'iterations': 22909}
frial 37: value = 1.064503e+01, params = {'newton_h': 1.0005276439975017, 'c1': 0.35495566132189404, 'a0': 1.6966904113314958, 'q': 0.6857387594623001, 'iterations': 36659}
frial 38: value = 1.064503e+01, params = {'newton_h': 1.628327232363436, 'c1': 0.10141395414681037, 'a0': 1.0776951625218651, 'q': 0.4641195916884134, 'iterations': 13867}
frial 39: value = 1.064503e+01, params = {'newton_h': 0.6811403263567353, 'c1': 0.14175064504241727, 'a0': 2.8661096970985143, 'q': 0.38910641002818835, 'iterations': 45576}
frial 40: value = 1.064503e+01, params = {'newton_h': 2.123489278029572, 'c1': 0.40821567577479784, 'a0': 2.62426893922758, 'q': 0.6119186579854621, 'iterations': 4679}
frial 41: value = 1.064503e+01, params = {'newton_h': 0.5624440768694022, 'c1': 0.3785635190690483, 'a0': 2.7618054238113716, 'q': 0.7081213254969954, 'iterations': 17538}
frial 42: value = 1.064503e+01, params = {'newton_h': 0.37971057856324497, 'c1': 0.4451380915806351, 'a0': 2.408637626433141, 'q': 0.7557552275950219, 'iterations': 9966}
frial 43: value = 1.064503e+01, params = {'newton_h': 0.8506327029039, 'c1': 0.48610263453532787, 'a0': 2.6193507681445767, 'q': 0.66116879038089251, 'iterations': 20237}
frial 44: value = 1.064503e+01, params = {'newton_h': 0.31291001792179135, 'c1': 0.40629452077574507, 'a0': 2.269202668047293, 'q': 0.7373409994752983, 'iterations': 14479}
frial 45: value = 1.064503e+01, params = {'newton_h': 1.7853005957592518, 'c1': 0.2798500132655806, 'a0': 2.900917881985669, 'q': 0.6917131294374461, 'iterations': 22532}
frial 46: value = 1.064503e+01, params = {'newton_h': 1.1355978889088867, 'c1': 0.19929352427161418, 'a0': 2.8194804427969475, 'q': 0.7974042680730929, 'iterations': 29382}
frial 47: value = 1.064503e+01, params = {'newton_h': 2.3903798966417655, 'c1': 0.47679729601365134, 'a0': 1.9120863863252449, 'q': 0.6664129309133379, 'iterations': 40116}
frial 48: value = 1.064503e+01, params = {'newton_h': 0.4607255557840009, 'c1': 0.355259717375934107, 'a0': 2.137909106661051, 'q': 0.3281890887059071, 'iterations': 15882}
frial 49: value = 1.064503e+01, params = {'newton_h': 0.6295366952316948, 'c1': 0.23075057040265007, 'a0': 2.6329145608689363, 'q': 0.6195366811326701, 'iterations': 9782}
=====
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Newton_armijo: f4

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Hyperparameter optimization (newton_armijo) for: f4 = x**2 + y**2 * np.sin(3 * x) * np.cos(2 * y)
-----
test hyperparameters:
newton_h = 1.912652
c1       = 0.247807
a0       = 2.961952
q        = 0.401318
iterations= 38430
test objective value: -2.766236e+02
-----
trial details:
Trial 0: value = 1.739792e+01, params = {'newton_h': 0.7041516740494138, 'c1': 0.32563667340580743, 'a0': 0.6553866075695126, 'q': 0.6268734105000293, 'iterations': 2750}
Trial 1: value = 2.948623e+00, params = {'newton_h': 1.644359874302552, 'c1': 0.19913207470934116, 'a0': 1.543531534813008, 'q': 0.40286887376820135, 'iterations': 10500}
Trial 2: value = 1.753076e+01, params = {'newton_h': 0.91070480895529167, 'c1': 0.4666410486329442, 'a0': 1.8869370450878452, 'q': 0.78318914956014, 'iterations': 47326}
Trial 3: value = -8.686285e+01, params = {'newton_h': 1.1668579596437125, 'c1': 0.2514057437570417, 'a0': 1.9303732180143882, 'q': 0.41632099439451226, 'iterations': 489}
Trial 4: value = 9.839893e+00, params = {'newton_h': 0.8767791137538768, 'c1': 0.3629566448434941, 'a0': 1.4060633740093773, 'q': 0.7111060098473567, 'iterations': 21638}
Trial 5: value = 6.883353e-01, params = {'newton_h': 1.4022953364488102, 'c1': 0.1946954604880086, 'a0': 2.1710405200168834, 'q': 0.34051373298640935, 'iterations': 6321}
Trial 6: value = -3.827229e-02, params = {'newton_h': 1.10029274177221369, 'c1': 0.16233473637442147, 'a0': 2.385086153342531, 'q': 0.60806973270800877, 'iterations': 2466}
Trial 7: value = 1.574005e+01, params = {'newton_h': 2.108571402401699, 'c1': 0.463593862378722, 'a0': 1.508202959357801, 'q': 0.6666157740332757, 'iterations': 47712}
Trial 8: value = 4.899088e-34, params = {'newton_h': 0.1096145124989898, 'c1': 0.389972516247119, 'a0': 2.344206506010557, 'q': 0.6905086875507866, 'iterations': 6129}
Trial 9: value = -3.827229e-02, params = {'newton_h': 2.626060613939361, 'c1': 0.36150589972524183, 'a0': 2.6615023451909634, 'q': 0.6387517376643869, 'iterations': 1596}
Trial 10: value = -2.766236e+02, params = {'newton_h': 1.912652492188014, 'c1': 0.247807865740384, 'a0': 2.961952066803932, 'q': 0.4813175727542016, 'iterations': 38430}
Trial 11: value = -2.654774e+00, params = {'newton_h': 1.941570173728915, 'c1': 0.25258070326014, 'a0': 2.999297491067142, 'q': 0.5010858783209053, 'iterations': 41002}
Trial 12: value = 1.594929e+01, params = {'newton_h': 2.9668027531325265, 'c1': 0.11462646235035201, 'a0': 0.8557865841422871, 'q': 0.4756288952849552, 'iterations': 3602}
Trial 13: value = 1.244367e+01, params = {'newton_h': 2.0671162927716957, 'c1': 0.276449808050848, 'a0': 1.1456065583826451, 'q': 0.427009345923162, 'iterations': 34121}
Trial 14: value = -6.187064e-01, params = {'newton_h': 1.4348255518334725, 'c1': 0.22742232465736895, 'a0': 2.91277075045746, 'q': 0.30993874686305786, 'iterations': 40528}
Trial 15: value = -3.827229e-02, params = {'newton_h': 0.37963217098083835, 'c1': 0.30374047979326013, 'a0': 1.9364069904542539, 'q': 0.5406460742966029, 'iterations': 498}
Trial 16: value = -3.827229e-02, params = {'newton_h': 2.415211319443195, 'c1': 0.12294046168864911, 'a0': 2.6203474144577914, 'q': 0.38339595654191444, 'iterations': 418}
Trial 17: value = 1.229304e+01, params = {'newton_h': 1.7240382289394485, 'c1': 0.24348798944948813, 'a0': 1.1583134434070805, 'q': 0.4523678225983343, 'iterations': 3182}
Trial 18: value = -1.449976e+02, params = {'newton_h': 1.227710681672573, 'c1': 0.1592886871619897, 'a0': 2.094392257562989, 'q': 0.546883114117452, 'iterations': 44308}
Trial 19: value = -3.827229e-02, params = {'newton_h': 1.815336384442421, 'c1': 0.1623846021880283, 'a0': 2.6759106763119407, 'q': 0.567084789640411, 'iterations': 38335}
Trial 20: value = -1.228255e-04, params = {'newton_h': 2.312397175647786, 'c1': 0.16727471176738, 'a0': 2.22570125324409, 'q': 0.522980974122205, 'iterations': 31504}
Trial 21: value = -9.394465e+00, params = {'newton_h': 1.2403060806064867, 'c1': 0.21048151041051302, 'a0': 2.0187955293074213, 'q': 0.4732016115791129, 'iterations': 4467}
Trial 22: value = -3.827229e-02, params = {'newton_h': 4.4203748953041241, 'c1': 0.22673440914061195, 'a0': 1.7030909707045745, 'q': 0.5707313701307520, 'iterations': 1776}
```

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Trial 23: value = -7.935310e+00, params = {'newton_h': 0.646717181017054, 'c1': 0.31938236460764224, 'a0': 2.4773785025684185, 'q': 0.367332639533442, 'iterations': 44347
Trial 24: value = -3.978033e-04, params = {'newton_h': 1.44175624154773, 'c1': 0.14535961186660562, 'a0': 2.0544571549666926, 'q': 0.41819102980236067, 'iterations': 3701
Trial 25: value = -6.187064e-01, params = {'newton_h': 1.6826269387804242, 'c1': 0.2669262182708481, 'a0': 1.689419301775752, 'q': 0.5100231546292289, 'iterations': 49546
Trial 26: value = -7.067550e-01, params = {'newton_h': 1.156968138748001, 'c1': 0.102329229529229457, 'a0': 1.3536494628769735, 'q': 0.4643988745662763, 'iterations': 2954
Trial 27: value = -7.148067e-04, params = {'newton_h': 1.890367657264982, 'c1': 0.21844569597267838, 'a0': 2.836338682302459, 'q': 0.35377294597644543, 'iterations': 4330
Trial 28: value = -9.614119e-04, params = {'newton_h': 0.9273716938432109, 'c1': 0.1781133437282737, 'a0': 2.181052817134362, 'q': 0.4398506182597078, 'iterations': 38829
Trial 29: value = -3.827229e-02, params = {'newton_h': 0.48579994194704346, 'c1': 0.35289479135377285, 'a0': 2.5155015018345646, 'q': 0.611020898828004, 'iterations': 235
Trial 30: value = 1.616467e+01, params = {'newton_h': 0.6962047314003599, 'c1': 0.302439166758004684, 'a0': 0.8361732903953047, 'q': 0.501270816079493, 'iterations': 19368
Trial 31: value = -1.145596e+02, params = {'newton_h': 1.2174956384369864, 'c1': 0.21387830672921135, 'a0': 1.998110112339365, 'q': 0.48892350548893965, 'iterations': 455
Trial 32: value = -3.827229e-02, params = {'newton_h': 1.3601835533653235, 'c1': 0.23811391862285908, 'a0': 1.7481921919312887, 'q': 0.40559362879240124, 'iterations': 468
Trial 33: value = -1.145596e+02, params = {'newton_h': 0.9322313015445054, 'c1': 0.18497962657132974, 'a0': 2.0546600317463035, 'q': 0.5433080824672246, 'iterations': 462
Trial 34: value = -2.310270e-08, params = {'newton_h': 0.9483442718954126, 'c1': 0.13490096809646362, 'a0': 2.26636438055342, 'q': 0.550282608205305, 'iterations': 34815}
Trial 35: value = -6.187064e-01, params = {'newton_h': 1.5901419131286483, 'c1': 0.2038164340203518, 'a0': 2.0712825368300427, 'q': 0.5807666918320903, 'iterations': 4657
Trial 36: value = 9.807113e+00, params = {'newton_h': 0.8117085373249593, 'c1': 0.18285797963038677, 'a0': 1.4108933235236313, 'q': 0.7812068208203709, 'iterations': 892}
Trial 37: value = -3.827229e-02, params = {'newton_h': 1.2953001794877796, 'c1': 0.14587801069819928, 'a0': 1.8742448333772792, 'q': 0.5370315591268319, 'iterations': 397
Trial 38: value = -3.827229e-02, params = {'newton_h': 1.032649787182155, 'c1': 0.2198011029137462, 'a0': 1.5458508157961206, 'q': 0.48893419365239493, 'iterations': 2754
Trial 39: value = 1.754590e+01, params = {'newton_h': 0.5237167977969182, 'c1': 0.18523757182685754, 'a0': 0.580271562047331, 'q': 0.6532011280301304, 'iterations': 4237
Trial 40: value = 9.307234e-01, params = {'newton_h': 1.711057277085966, 'c1': 0.1970199545547151, 'a0': 2.4182421889675574, 'q': 0.5994377143912404, 'iterations': 46894}
Trial 41: value = -8.686285e+01, params = {'newton_h': 1.218836559060469, 'c1': 0.26194313625186993, 'a0': 1.9275757270053377, 'q': 0.39102416869892964, 'iterations': 490
Trial 42: value = -8.095150e-05, params = {'newton_h': 0.7975606377617401, 'c1': 0.28309511791807307, 'a0': 2.1112023943275724, 'q': 0.4591063698321956, 'iterations': 45
Trial 43: value = -3.827229e-02, params = {'newton_h': 1.02706469989473963, 'c1': 0.239679167111391105, 'a0': 1.631800341717751, 'q': 0.5244434034226231, 'iterations': 4281
Trial 44: value = -3.827229e-02, params = {'newton_h': 1.4978905252219497, 'c1': 0.19657259177497455, 'a0': 2.304423383840373, 'q': 0.4863258769075116, 'iterations': 4803
Trial 45: value = 1.753076e+01, params = {'newton_h': 1.3482499626504874, 'c1': 0.49411860813334846, 'a0': 1.8557844837252333, 'q': 0.4354369145437492, 'iterations': 5000
Trial 46: value = 1.291129e+01, params = {'newton_h': 0.21235378622123102, 'c1': 0.39147115327111415, 'a0': 1.958681279088697, 'q': 0.5652363571283776, 'iterations': 1254
Trial 47: value = 9.812683e+00, params = {'newton_h': 2.053599179619394, 'c1': 0.2283255334601729, 'a0': 1.409407058047861, 'q': 0.3339247541417089, 'iterations': 40574}
Trial 48: value = -2.912705e+01, params = {'newton_h': 1.036449770602775, 'c1': 0.25730554059442456, 'a0': 2.800312767358953, 'q': 0.61929776650824368, 'iterations': 37985
Trial 49: value = -3.827229e-02, params = {'newton_h': 1.5222052500009755, 'c1': 0.290215111330405275, 'a0': 2.594801234558718, 'q': 0.5898716624710927, 'iterations': 3315
;=====
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Newton_armijo: multimodal_f

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=====
Hyperparameter optimization (newton_armijo) for: multimodal_f = 20 + x^2 + y^2 - 10*cos(2*pi*x) - 10*cos(2*pi*y)
-----
Best hyperparameters:
  newton_h = 2.485039
  c1       = 0.284883
  a0       = 1.753288
  q        = 0.486502
  iterations = 14183
Best objective value: 2.885355e+01
-----
Trial details:
Trial 0: value = 2.885355e+01, params = {'newton_h': 2.55093656430408594, 'c1': 0.33778506080495141, 'a0': 1.87532866480246505, 'q': 0.5029623294762791, 'iterations': 19424
Trial 1: value = 2.885355e+01, params = {'newton_h': 2.982253768286198, 'c1': 0.32217746662071, 'a0': 1.2456012680285737, 'q': 0.5983500239975673, 'iterations': 44732}
Trial 2: value = 2.885355e+01, params = {'newton_h': 2.906350767239068, 'c1': 0.47429900876157544, 'a0': 2.7593121468405104, 'q': 0.5562139971295416, 'iterations': 20595
Trial 3: value = 2.885355e+01, params = {'newton_h': 2.077060902043236, 'c1': 0.1403114680711066, 'a0': 1.3327231076144925, 'q': 0.45371492878877223, 'iterations': 30377
Trial 4: value = 2.885355e+01, params = {'newton_h': 1.8285252830878116, 'c1': 0.36688816130533763, 'a0': 2.2760942411397487, 'q': 0.30718397484159354, 'iterations': 121
Trial 5: value = 2.885355e+01, params = {'newton_h': 0.4467567676534375, 'c1': 0.34181466312060027, 'a0': 2.551433149974699, 'q': 0.6457558323349921, 'iterations': 25435
Trial 6: value = 2.885355e+01, params = {'newton_h': 2.010954908580595, 'c1': 0.47677580563960086, 'a0': 0.9227135688961368, 'q': 0.3866975015158466, 'iterations': 2969}
Trial 7: value = 2.885355e+01, params = {'newton_h': 1.6900876601409176, 'c1': 0.47135281079782065, 'a0': 1.7997190378989072, 'q': 0.438149281966429, 'iterations': 2878}
Trial 8: value = 2.885355e+01, params = {'newton_h': 2.2010425882009566, 'c1': 0.31956727221031267, 'a0': 2.3283612239904783, 'q': 0.42443560715647524, 'iterations': 273
Trial 9: value = 2.885355e+01, params = {'newton_h': 1.3573305673138671, 'c1': 0.46333142463004395, 'a0': 1.9983170802778494, 'q': 0.31781371214856396, 'iterations': 688
Trial 10: value = 2.885355e+01, params = {'newton_h': 1.033379568168149, 'c1': 0.19046636674640272, 'a0': 1.4479498761282121, 'q': 0.7103999933651786, 'iterations': 15556
Trial 11: value = 2.885355e+01, params = {'newton_h': 2.945542320218782, 'c1': 0.24797147719911672, 'a0': 0.5257016109600448, 'q': 0.5704697151975666, 'iterations': 48457
Trial 12: value = 2.885355e+01, params = {'newton_h': 2.5749100027054634, 'c1': 0.3850240674427541, 'a0': 1.4232093247017303, 'q': 0.7948597330955514, 'iterations': 39503
Trial 13: value = 2.885355e+01, params = {'newton_h': 2.5281183318049507, 'c1': 0.25326247594339035, 'a0': 1.0378874599753307, 'q': 0.6180720541156170, 'iterations': 3036
Trial 14: value = 2.885355e+01, params = {'newton_h': 2.4850388651872706, 'c1': 0.2848830031339652, 'a0': 1.753287751129455, 'q': 0.48650192883840543, 'iterations': 14183
Trial 15: value = 2.885355e+01, params = {'newton_h': 2.3976785467225596, 'c1': 0.25708654012124355, 'a0': 1.9060941516868728, 'q': 0.48324524603990103, 'iterations': 125
Trial 16: value = 2.885355e+01, params = {'newton_h': 1.14008044985938516, 'c1': 0.25437960888220074, 'a0': 2.9606036027761524, 'q': 0.5052522865956675, 'iterations': 1114
Trial 17: value = 2.885355e+01, params = {'newton_h': 2.358117641548795, 'c1': 0.20485189227180675, 'a0': 2.062752130142022, 'q': 0.49785027575519034, 'iterations': 11554
Trial 18: value = 2.885355e+01, params = {'newton_h': 0.7632389830861757, 'c1': 0.11629098730269971, 'a0': 1.5141983465646902, 'q': 0.36376524700617696, 'iterations': 178
Trial 19: value = 2.885355e+01, params = {'newton_h': 1.494720813198155, 'c1': 0.3998905250458127, 'a0': 1.625825097682185, 'q': 0.6971113754110732, 'iterations': 9630}
Trial 20: value = 2.885355e+01, params = {'newton_h': 1.8832811682181057, 'c1': 0.2875811336937267, 'a0': 2.296824094615092, 'q': 0.4823570588591217, 'iterations': 32162}
Trial 21: value = 2.885355e+01, params = {'newton_h': 1.465897561672647, 'c1': 0.4103737005541313, 'a0': 1.6323284711061645, 'q': 0.7331805359136438, 'iterations': 10732}
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Trial 23: value = 2.885355e+01, params = {'newton_h': 0.11197896869621138, 'c1': 0.2817208266463103, 'a0': 2.069790786219373, 'q': 0.5380757218525003, 'iterations': 8247}
Trial 24: value = 2.885355e+01, params = {'newton_h': 2.2681869296076647, 'c1': 0.17725938038654399, 'a0': 1.6686655922155684, 'q': 0.7971942075711674, 'iterations': 22075}
Trial 25: value = 2.885355e+01, params = {'newton_h': 2.6958722967632145, 'c1': 0.2273419732450272, 'a0': 1.1218542328474722, 'q': 0.3871617111655771, 'iterations': 7343}
Trial 26: value = 2.885355e+01, params = {'newton_h': 1.153568293738615, 'c1': 0.2870947011123374, 'a0': 0.8333085208539106, 'q': 0.5317908228028798, 'iterations': 13262}
Trial 27: value = 2.885355e+01, params = {'newton_h': 2.3479128973213976, 'c1': 0.380965843103883, 'a0': 1.9393085710148732, 'q': 0.7312961359860815, 'iterations': 5052}
Trial 28: value = 2.885355e+01, params = {'newton_h': 1.3663667673616793, 'c1': 0.4356602697588303, 'a0': 2.1691778019628396, 'q': 0.598402746148295, 'iterations': 21340}
Trial 29: value = 2.885355e+01, params = {'newton_h': 2.6839579307794263, 'c1': 0.3389809258483138, 'a0': 1.8358626678696672, 'q': 0.474527760960675, 'iterations': 17443}
Trial 30: value = 2.885355e+01, params = {'newton_h': 0.8971710826870646, 'c1': 0.22127115525232965, 'a0': 2.4997410879772844, 'q': 0.6750427489781631, 'iterations': 136}
Trial 31: value = 2.885355e+01, params = {'newton_h': 1.34949774931102, 'c1': 0.398097332869168, 'a0': 1.7282942861569066, 'q': 0.7562519875157447, 'iterations': 8886}
Trial 32: value = 2.885355e+01, params = {'newton_h': 1.502869544024697, 'c1': 0.4181490309529412, 'a0': 1.6027287385917848, 'q': 0.7058343008283086, 'iterations': 11383}
Trial 33: value = 2.885355e+01, params = {'newton_h': 1.8278641547625258, 'c1': 0.3170127683545808, 'a0': 1.2637892581429275, 'q': 0.6262292220892405, 'iterations': 15495}
Trial 34: value = 2.885355e+01, params = {'newton_h': 1.5887502331134378, 'c1': 0.3608630900675834, 'a0': 1.5816120727175917, 'q': 0.7644392812655164, 'iterations': 18295}
Trial 35: value = 2.885355e+01, params = {'newton_h': 2.1448788837819146, 'c1': 0.40995657948162495, 'a0': 1.893925070189158, 'q': 0.5815283022620236, 'iterations': 5150}
Trial 36: value = 2.885355e+01, params = {'newton_h': 1.9896376330614194, 'c1': 0.3615266972276974, 'a0': 1.3446345154686639, 'q': 0.5240661814620785, 'iterations': 23312}
Trial 37: value = 2.885355e+01, params = {'newton_h': 1.78159751323025, 'c1': 0.4942052176255365, 'a0': 1.1636679333969244, 'q': 0.45735084361402445, 'iterations': 9895}
Trial 38: value = 2.885355e+01, params = {'newton_h': 2.7941478286142534, 'c1': 0.4440829068927653, 'a0': 2.1462822349220465, 'q': 0.4154919536583183, 'iterations': 12957}
Trial 39: value = 2.885355e+01, params = {'newton_h': 0.6391593017227641, 'c1': 0.3090962292665295, 'a0': 2.450298039187654, 'q': 0.6766482477431669, 'iterations': 26940}
Trial 40: value = 2.885355e+01, params = {'newton_h': 2.429681626894366, 'c1': 0.26789948110154954, 'a0': 1.668388402966711, 'q': 0.5567714144245817, 'iterations': 5394}
Trial 41: value = 2.885355e+01, params = {'newton_h': 1.4620729496740617, 'c1': 0.3263176136833184, 'a0': 1.257239994634696, 'q': 0.6408777109168151, 'iterations': 16378}
Trial 42: value = 2.885355e+01, params = {'newton_h': 1.9334094903653236, 'c1': 0.3037377647240371, 'a0': 1.3539755508258011, 'q': 0.6246082582508464, 'iterations': 20259}
Trial 43: value = 2.885355e+01, params = {'newton_h': 2.1147744252963987, 'c1': 0.15917490512893853, 'a0': 1.830052973883088, 'q': 0.70382124777161205, 'iterations': 14342}
Trial 44: value = 2.885355e+01, params = {'newton_h': 1.767531797901555, 'c1': 0.4530266230517032, 'a0': 0.7675201814061519, 'q': 0.7530727566733996, 'iterations': 2746}
Trial 45: value = 2.885355e+01, params = {'newton_h': 1.208835553247351, 'c1': 0.34595602265775244, 'a0': 1.5028368159562766, 'q': 0.6535210320735145, 'iterations': 9795}
Trial 46: value = 2.885355e+01, params = {'newton_h': 1.753270975602731, 'c1': 0.3023612942803898, 'a0': 1.0086315545371516, 'q': 0.731295912679349, 'iterations': 16156}
Trial 47: value = 2.885355e+01, params = {'newton_h': 2.031569767731425, 'c1': 0.23465557790852132, 'a0': 1.271742670448319, 'q': 0.6089350031808343, 'iterations': 13594}
Trial 48: value = 2.885355e+01, params = {'newton_h': 2.4971486044335345, 'c1': 0.4001492739298884, 'a0': 1.4519962043877306, 'q': 0.508325840100417, 'iterations': 19958}
Trial 49: value = 2.885355e+01, params = {'newton_h': 1.2602018792370817, 'c1': 0.2716757185763386, 'a0': 1.948175085722673, 'q': 0.6860227825014633, 'iterations': 29840}
=====
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Newton_armijo: noisy_multimodal_f

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=====
Hyperparameter optimization (newton_armijo) for: noisy_multimodal_f = ∑ sin(m·x)*cos(n·y), m = 1..M + N(0, σ)
-----
Best hyperparameters:
newton_h   = 1.863075
c1         = 0.373118
a0         = 2.732207
q          = 0.663880
iterations= 9604
Best objective value: -2.400265e+00
-----
Trial details:
Trial 0: value = -1.429778e-01, params = {'newton_h': 1.8952425217082258, 'c1': 0.4665347410224089, 'a0': 2.603084013333849, 'q': 0.7853373834108239, 'iterations': 18196}
Trial 1: value = 1.227986e+00, params = {'newton_h': 0.3797659010877727, 'c1': 0.4727861774735109, 'a0': 1.181181130297861, 'q': 0.619182909617447, 'iterations': 5434}
Trial 2: value = 1.227986e+00, params = {'newton_h': 2.084209301813834, 'c1': 0.3151563999313195, 'a0': 1.3852517948861915, 'q': 0.565044104256257, 'iterations': 15817}
Trial 3: value = 1.227986e+00, params = {'newton_h': 2.1024139194017497, 'c1': 0.18424488408188813, 'a0': 1.487637606402956, 'q': 0.4177625243497082, 'iterations': 5775}
Trial 4: value = 1.478379e+00, params = {'newton_h': 2.7561742419532553, 'c1': 0.17542357010089316, 'a0': 0.6423240514831042, 'q': 0.6321413768339686, 'iterations': 4318}
Trial 5: value = 1.031418e+00, params = {'newton_h': 1.6113245484310892, 'c1': 0.47614938714107113, 'a0': 1.1066697166993267, 'q': 0.64951267134491, 'iterations': 3887}
Trial 6: value = 1.478379e+00, params = {'newton_h': 0.9583867898391115, 'c1': 0.11512914759662994, 'a0': 0.8659564425122628, 'q': 0.4751552740632814, 'iterations': 48824}
Trial 7: value = 1.478379e+00, params = {'newton_h': 2.2263027517960405, 'c1': 0.25975467564949745, 'a0': 0.6700984279702211, 'q': 0.49066979744049816, 'iterations': 308}
Trial 8: value = -1.468189e+00, params = {'newton_h': 1.967336933640624, 'c1': 0.419838040291351543, 'a0': 2.8093675329977126, 'q': 0.6455705465174963, 'iterations': 1806}
Trial 9: value = 1.647076e+00, params = {'newton_h': 1.566393190006229, 'c1': 0.37001439188221247, 'a0': 0.9406529852546631, 'q': 0.431122076170778, 'iterations': 37163}
Trial 10: value = -3.395457e-01, params = {'newton_h': 2.9188210887907258, 'c1': 0.38402574704701753, 'a0': 2.864624166662575, 'q': 0.7603646252063468, 'iterations': 174}
Trial 11: value = 1.227986e+00, params = {'newton_h': 2.858587016850375, 'c1': 0.30516495319206307, 'a0': 2.997716060454603, 'q': 0.7074463266127268, 'iterations': 17094}
Trial 12: value = 1.227986e+00, params = {'newton_h': 2.602870086449267, 'c1': 0.39097734017577024, 'a0': 2.1297309391652255, 'q': 0.71544476481857132, 'iterations': 1170}
Trial 13: value = 1.227986e+00, params = {'newton_h': 1.0239300607785886, 'c1': 0.3165106946073536, 'a0': 2.238821194211781, 'q': 0.32953257664436475, 'iterations': 2646}
Trial 14: value = 1.227986e+00, params = {'newton_h': 2.4796188162441384, 'c1': 0.41697104367696497, 'a0': 2.991643197740289, 'q': 0.7061288483800028, 'iterations': 10964}
Trial 15: value = 1.227986e+00, params = {'newton_h': 2.975271731244484, 'c1': 0.2666892518916449, 'a0': 2.580537240475352, 'q': 0.7271570708636925, 'iterations': 123}
Trial 16: value = 1.227986e+00, params = {'newton_h': 1.1800208746009617, 'c1': 0.42212377877367485, 'a0': 2.002879201528654, 'q': 0.680352152842189, 'iterations': 21333}
Trial 17: value = 1.227986e+00, params = {'newton_h': 0.30333204385887114, 'c1': 0.3288271192186865, 'a0': 2.607342245917651, 'q': 0.561104832410305, 'iterations': 26221}
Trial 18: value = 1.227986e+00, params = {'newton_h': 2.3874309451773463, 'c1': 0.49988658757021653, 'a0': 1.809978309643652, 'q': 0.5981427601998835, 'iterations': 10946}
Trial 19: value = 1.106276e+00, params = {'newton_h': 0.7081443684454851, 'c1': 0.35125688713982084, 'a0': 2.4035072736708525, 'q': 0.7519755099259482, 'iterations': 342}
Trial 20: value = -1.468189e+00, params = {'newton_h': 1.8403723042643771, 'c1': 0.4174784567067388, 'a0': 2.755202461132099, 'q': 0.6608045029191126, 'iterations': 244}
Trial 21: value = -3.395457e-01, params = {'newton_h': 1.7957605997390154, 'c1': 0.4382108359330733, 'a0': 2.798460516018381, 'q': 0.6704666617460586, 'iterations': 1397}
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Trial 23: value = -1.271621e+00, params = {'newton_h': 1.3122029487165405, 'c1': 0.4350829534511814, 'a0': 2.3527229881533738, 'q': 0.6878320774562068, 'iterations': 839}
Trial 24: value = -3.395457e-01, params = {'newton_h': 1.3300382667310506, 'c1': 0.2726520649339572, 'a0': 2.7125860804462834, 'q': 0.5237289002177827, 'iterations': 666}
Trial 25: value = 1.227986e+00, params = {'newton_h': 1.8131072555547179, 'c1': 0.34579352589539997, 'a0': 2.4483025741742495, 'q': 0.601832335121946, 'iterations': 8713}
Trial 26: value = 1.227986e+00, params = {'newton_h': 0.639620382630143, 'c1': 0.41430584378176827, 'a0': 1.8872393033291437, 'q': 0.7408720983727428, 'iterations': 4540}
Trial 27: value = -1.429778e-01, params = {'newton_h': 1.4289202012302964, 'c1': 0.49777006078191965, 'a0': 2.7980385569505906, 'q': 0.6596787030640809, 'iterations': 13}
Trial 28: value = 1.227986e+00, params = {'newton_h': 1.961504917666594, 'c1': 0.4510943611552721, 'a0': 2.189595787422284, 'q': 0.7988275356572263, 'iterations': 7145}
Trial 29: value = -1.271621e+00, params = {'newton_h': 1.691548326302547, 'c1': 0.40378239533815297, 'a0': 2.5709758517611236, 'q': 0.76429493936223682, 'iterations': 221}
Trial 30: value = 1.227986e+00, params = {'newton_h': 2.2822805767983967, 'c1': 0.4546045082495297, 'a0': 1.6375968513254402, 'q': 0.7026499217916685, 'iterations': 2941}
Trial 31: value = -1.429778e-01, params = {'newton_h': 1.32898811733636, 'c1': 0.43116279352377795, 'a0': 2.3769693030284116, 'q': 0.6866567828377292, 'iterations': 7947}
Trial 32: value = -1.271621e+00, params = {'newton_h': 1.1674087857204876, 'c1': 0.46905922868156336, 'a0': 2.6630891627899356, 'q': 0.6307191228951776, 'iterations': 82}
Trial 33: value = 9.934283e-02, params = {'newton_h': 1.983075952999375, 'c1': 0.36431043281882025, 'a0': 2.8479133863434916, 'q': 0.598580503990675, 'iterations': 1409}
Trial 34: value = 1.227986e+00, params = {'newton_h': 1.471646339529155, 'c1': 0.2948871406682876, 'a0': 2.480853154759687, 'q': 0.6532617591147664, 'iterations': 4738}
Trial 35: value = 1.227986e+00, params = {'newton_h': 1.7515975970696944, 'c1': 0.4414745757660061, 'a0': 2.2508093371642457, 'q': 0.576926417293747, 'iterations': 20474}
Trial 36: value = 1.128836e+00, params = {'newton_h': 0.7684771754674722, 'c1': 0.4703233742433015, 'a0': 2.707428075538388, 'q': 0.729917948342529, 'iterations': 2968}
Trial 37: value = 1.225411e+00, params = {'newton_h': 2.0459964786299714, 'c1': 0.22679503379143223, 'a0': 2.9349446419880643, 'q': 0.6327121155800586, 'iterations': 616}
Trial 38: value = 1.227986e+00, params = {'newton_h': 2.165706292832592, 'c1': 0.40917708158728294, 'a0': 2.341421209751737, 'q': 0.5275628243821916, 'iterations': 10045}
Trial 39: value = 1.227986e+00, params = {'newton_h': 0.950397516258982, 'c1': 0.3442182536860993, 'a0': 2.092048640951576, 'q': 0.6865199723774092, 'iterations': 13382}
Trial 40: value = 1.227986e+00, params = {'newton_h': 1.22426137990265917, 'c1': 0.118391771635797, 'a0': 1.498409275499318, 'q': 0.7701354686327743, 'iterations': 2111}
Trial 41: value = -3.395457e-01, params = {'newton_h': 1.1006906201335631, 'c1': 0.4758361327765751, 'a0': 2.6685108722195183, 'q': 0.623619253656482, 'iterations': 6720}
Trial 42: value = -2.400265e+00, params = {'newton_h': 1.5705003724047915, 'c1': 0.4618689379854526, 'a0': 2.7485480505541657, 'q': 0.6393508818652517, 'iterations': 901}
Trial 43: value = 1.209757e+00, params = {'newton_h': 1.653611246124048, 'c1': 0.39717583480192487, 'a0': 2.5162888121454645, 'q': 0.6968735208737388, 'iterations': 4745}
Trial 44: value = -3.395457e-01, params = {'newton_h': 1.526808942963142, 'c1': 0.45918995941095897, 'a0': 2.8663786397722184, 'q': 0.638431793382704, 'iterations': 154}
Trial 45: value = -2.400265e+00, params = {'newton_h': 1.8630749823711479, 'c1': 0.3731175198437481, 'a0': 2.732073887815512, 'q': 0.6638797787693351, 'iterations': 960}
Trial 46: value = -3.395457e-01, params = {'newton_h': 1.915050051929522, 'c1': 0.36687708691583104, 'a0': 2.753266471043427, 'q': 0.5335582400335657, 'iterations': 261}
Trial 47: value = 9.934283e-02, params = {'newton_h': 1.866791964035825, 'c1': 0.37331555359449714, 'a0': 1.2484804488137313, 'q': 0.5864649142917167, 'iterations': 4030}
Trial 48: value = -3.395457e-01, params = {'newton_h': 1.5944559057532577, 'c1': 0.40533656072556774, 'a0': 2.9217806949803693, 'q': 0.658565173762832, 'iterations': 488}
Trial 49: value = 1.031418e+00, params = {'newton_h': 2.1058611870906367, 'c1': 0.38686511275571883, 'a0': 2.9916388688953686, 'q': 0.3263173844693075, 'iterations': 299}
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Newton results:

Function	Method	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
f1 = x ² + y ²	Newton-CG	x = -1.8184686573248843e-15, y = -7.273874589594868e-16	x = -5.686801612204206e-17, y = 3.273233054709359e-17	1180	295	59
f3 = x ² + B·xy + y ² , where -2 < B < 2	Newton-CG	x = -4.768427605456895e-13, y = 1.9529044578505176e-12	x = -1.7919247447914917e-16, y = 2.269981893268966e-16	1124	281	57
f4 = x ² + y ² ·sin(3x)·cos(2y)	Newton-CG	x = -0.177556086096979, y = 0.5383618076832533	x = -1.0471975511427205, y = -2.2893973907188583	664	166	34
ackley_f = -20·exp(-0.2·√((x ² +y ²)/2)) - exp((cos(2πx)+cos(2πy))/2) + 20 + e	KvasiNewton (not converged)	–	x = 4.97684384945899, y = 1.9907648756230907	544	136	28
multimodal_f = 20 + x ² + y ² - 10·cos(2πx) - 10·cos(2πy)	KvasiNewton (not converged)	–	x = 4.974691390776108, y = 1.9899122337939812	824	206	42
noisy_multimodal_f = ∑ sin(m·x)·cos(m·y) + N(0,σ)	KvasiNewton (not converged)	–	x = 4.382837394918323, y = 2.5676389841440005	684	171	35

Newton_armijo results:

Function	Method	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
f1 = x ² + y ²	Newton-CG	x = -1.8184686573248843e-15, y = -7.273874589594868e-16	x = 0.0, y = 0.0	73	16	10
f3 = x ² + B·xy + y ² (with -2 < B < 2)	Newton-CG	x = -4.768427605456895e-13, y = 1.9529044578505176e-12	x = -3.308722450212111e-24, y = 0.0	119	26	16
f4 = x ² + y ² ·sin(3x)·cos(2y)	Newton-CG	x = -0.177556086096979, y = 0.5383618076832533	x = -0.1775911332234026, y = 0.538436993398883	407	85	67
ackley_f (-20·exp(-0.2·√((x ² +y ²)/2)) - exp((cos(2πx)+cos(2πy))/2) + 20 + e)	KvasiNewton (not converged)	–	x = 4.976843849392612, y = 1.9907648759478566	265	56	42
multimodal_f (20 + x ² + y ² - 10·cos(2πx) - 10·cos(2πy))·*	KvasiNewton (not converged)	–	x = 4.974691390900148, y = 1.9899122339537112	183	35	43
noisy_multimodal_f (∑ sin(m·x)·cos(m·y), m = 1..M + N(0, σ))·*	KvasiNewton (not converged)	–	x = 4.382837065945984, y = 2.5676387630263635	44,002	10,000	4,002

Nelder-Mid results:

Function	Method	SciPy Results
f1 = $x^2 + y^2$	Nelder-Mid	x = -2.664083899945463e-06, y = 2.1972263493949345e-05
f3 = $x^2 + B \cdot xy + y^2$ (with $-2 < B < 2$)	Nelder-Mid	x = 1.3318560856435198e-05, y = -2.149278762080329e-05
f4 = $x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	Nelder-Mid	x = -0.5125869365088374, y = 3.2891431610545343
ackley_f $(-20 \exp(-0.2 \sqrt{((x^2+y^2)/2)}) - \exp((\cos(2\pi x) + \cos(2\pi y))/2) + 20 + e)$	Nelder-Mid	x = 4.976795780317502, y = 1.9907713246158665
multimodal_f $(20 + x^2 + y^2 - 10 \cos(2\pi x) - 10 \cos(2\pi y))$	Nelder-Mid	x = 4.974658277356411, y = 1.9899280436352451
noisy_multimodal_f $(\sum \sin(m \cdot x) \cdot \cos(m \cdot y), m = 1..M + N(0, \sigma))$	Nelder-Mid	x = 5.8707772188400895, y = 2.2276621880859677

BFGS results:

Function	Method	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
f1 = $x^2 + y^2$	BFGS	x = -9.240271658939037e-08, y = 2.799203131509387e-07	x = 6.825872908091658e-18, y = 2.7303482533379894e-18	36	6	17
f3 = $x^2 + B \cdot xy + y^2$ (with $-2 < B < 2$)	BFGS	x = -5.341302798064998e-09, y = -3.162699161650024e-09	x = 2.8621171205784337e-16, y = -3.022747024181593e-16	7 172	752	4 169
f4 = $x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	BFGS (not converged)	–	x = -0.17759111322677656, y = 0.5384369939853243	65 209	2 001	57 209
ackley_f $(-20 \exp(-0.2 \sqrt{((x^2+y^2)/2)}) - \exp((\cos(2\pi x) + \cos(2\pi y))/2) + 20 + e)$	BFGS	x = 4.97684383587883, y = 1.9907648714033925	x = 1.958919717662159, y = 2.103214081234854e-09	69 268	2 001	61 268
multimodal_f $(20 + x^2 + y^2 - 10 \cos(2\pi x) - 10 \cos(2\pi y))$	BFGS	x = 4.974691392230302, y = 1.9899122292606068	x = 1.98991223395392, y = 0.9949586389016928	96 636	2 001	88 636
noisy_multimodal_f $(\sum \sin(m \cdot x) \cdot \cos(m \cdot y), m = 1..M + N(0, \sigma))$	BFGS	x = 6.857138849679963, y = 1.2412447421959505	x = 6.857138976446866, y = 1.241244740935296	383	31	264

Newton results:

Function	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
f1 = $x^2 + y^2$	x = -1.8184686573248843e-15, y = -7.273874589594868e-16	x = -5.686801612204206e-17, y = 3.273233054709359e-17	1180	295	59
f1_1 = $(x + 2)^2 + y^2$	x = -1.999999999313849, y = 1.9604577756604315e-10	x = -2.0000000000000004, y = 1.3083288963175647e-29	200 000	50 000	10 000
f1_2 = $x^2 + (y - 3)^2$	x = 1.4369200486601792e-11, y = 2.999999999971263	x = 9.055132664049583e-17, y = 3.0	1124	281	57
f1_3 = $(x - 2)^2 + (y + 1)^2$	x = 1.9999999995692785, y = -1.0000000004307237	x = 2.0, y = -1.0	564	141	29
f3 = $x^2 + B \cdot xy + y^2$ ($-2 < B < 2$)	x = -4.768427605456895e-13, y = 1.9529044578505176e-12	x = -1.7912947447914917e-16, y = 2.269981893268966e-16	1124	281	57
f4 = $x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	x = -0.177556086096979, y = 0.5383618076832533	x = -1.0471975511427205, y = -2.2893973907188583	664	166	34
ackley_f (Ackley function)	–	x = 4.97684384945899, y = 1.9907648756230907	544	136	28
multimodal_f (Multimodal function)	–	x = 4.974691390776108, y = 1.9899122337939812	824	206	42
noisy_multimodal_f (Noisy multimodal function)	–	x = 4.382837394918323, y = 2.5676389841440805	684	171	35

Newton_armijo results:

Function	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
$f1 = x^2 + y^2$	$x = -1.8184686573248843e-15, y = -7.273874589594868e-16$	$x = 0.0, y = 0.0$	73	16	10
$f1_1 = (x + 2)^2 + y^2$	$x = -1.999999999313849, y = 1.9604577756604315e-10$	$x = -2.0, y = 0.0$	119	26	16
$f1_2 = x^2 + (y - 3)^2$	$x = 1.4369200486601792e-11, y = 2.999999999971263$	$x = 2.3161057151484775e-23, y = 3.0$	96	21	13
$f1_3 = (x - 2)^2 + (y + 1)^2$	$x = 1.9999999995692785, y = -1.0000000004307237$	$x = 2.0, y = -1.0$	119	26	16
$f3 = x^2 + B \cdot x \cdot y + y^2$ ($-2 < B < 2$)	$x = -4.768427605456895e-13, y = 1.9529044578505176e-12$	$x = -3.308722450212111e-24, y = 0.0$	119	26	16
$f4 = x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	$x = -0.177556086096979, y = 0.5383618076832533$	$x = -0.17759111332234026, y = 0.538436993398883$	407	85	67
ackley_f (Ackley function)	–	$x = 4.976843849392612, y = 1.9907648759478556$	265	56	42
multimodal_f (Multimodal function)	–	$x = 4.974691390900148, y = 1.9899122339537112$	183	35	43
noisy_multimodal_f (Noisy multimodal function)	–	$x = 4.3828367344824315, y = 2.5676385954517396$	220 002	50 000	20 002

Nelder-Mid results:

Function	Result (x, y)
$f1 = x^2 + y^2$	(-2.664083899945463e-06, 2.1972263493949345e-05)
$f1_1 = (x + 2)^2 + y^2$	(-2.0000037105214226, 2.702663182143226e-05)
$f1_2 = x^2 + (y - 3)^2$	(6.939248329365828e-06, 2.9999797701324966)
$f1_3 = (x - 2)^2 + (y + 1)^2$	(1.9999967051243135, -1.0000368690121255)
$f3 = x^2 + B \cdot x \cdot y + y^2$ ($-2 < B < 2$)	(1.3318560856435198e-05, -2.149278762080329e-05)
$f4 = x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	(-0.5125869365088374, 3.2891431610545343)
ackley_f (Ackley function)	(4.976795780317502, 1.9907713246158665)
multimodal_f (Multimodal function)	(4.974658277356411, 1.9899280436352451)
noisy_multimodal_f (Noisy multimodal function)	(5.8707772188400895, 2.2276621880859677)

BFGS results:

Function	SciPy Results	Results	Function Evaluations	Gradient Evaluations	Iterations
$f1 = x^2 + y^2$	$x = -9.240271658939037e-08, y = 2.799203131509387e-07$	$x = 6.825872908091658e-18, y = 2.7303482533379894e-18$	36	6	17
$f1_1 = (x + 2)^2 + y^2$	$x = -2.00000059157997, y = 2.3412493019847602e-06$	$x = -2.0, y = 8.217936815318373e-18$	36	6	17
$f1_2 = x^2 + (y - 3)^2$	$x = 1.8257681100308787e-07, y = 3.0000004712101993$	$x = 4.426045063671209e-17, y = 3.0$	36	6	17
$f1_3 = (x - 2)^2 + (y + 1)^2$	$x = 1.9999999906707564, y = -1.0000000093292438$	$x = 2.0, y = -1.0$	45	7	22
$f3 = x^2 + B \cdot x \cdot y + y^2$ ($-2 < B < 2$)	$x = -5.341302798064998e-09, y = -3.162699161650024e-09$	$x = 2.8621171205784337e-16, y = -3.022747024181593e-16$	7 172	752	4 169
$f4 = x^2 + y^2 \cdot \sin(3x) \cdot \cos(2y)$	–	$x = -0.17759111322677656, y = 0.5384369939853243$	329 209	10 001	289 209
ackley_f (Ackley function)	$x = 4.97684383587883, y = 1.9907648714033925$	$x = 1.958919717662159, y = 1.911633030879068e-09$	349 268	10 001	309 268
multimodal_f (Multimodal function)	$x = 4.974691392230302, y = 1.9899122292606068$	$x = 1.98991223395392, y = 0.9949586389016928$	488 636	10 001	448 636
noisy_multimodal_f (Noisy multimodal function)	$x = 6.857138849679963, y = 1.2412447421959505$	$x = 6.857138976446866, y = 1.241244740935296$	383	31	264

Вывод

В ходе выполнения лабораторной работы были изучены и сравнены четыре метода оптимизации (нулевого и первого порядка) на примерах следующих функций:

Ackley (ackley_f),

Test-функции №4 (f4),

Мультимодальная (multimodal_f),

Шумная мультимодальная (noisy_multimodal_f).

Исследование проводилось от единой начальной точки $(x_0, y_0) = (1.0, 1.0)$ для всех методов. В качестве критерия останова в каждом случае использовалось либо • норма градиента $\|\nabla f\| < 10^{-6}$, • либо достижение максимального числа итераций (100).

Гиперпараметры и стратегия выбора шага для методов первого порядка были одинаковыми:

линейный поиск по Армихо с параметрами $c_1 = 10^{-4}$, $a_0 = 1.0$, $q = 0.5$

начальный шаг в методе Бroyдена–Флетчера–Голдфарба–Шенно (BFGS) выбирался как $p_0 = -\nabla f / \|\nabla f\|$

для метода Ньютона при вырожденном Гессиане направление брали как $p = -\nabla f$

Метод Нельдера–Мида (нулевого порядка) работал без градиентов, использовал стандартную стратегию отражения/расширения/сжатия симплекса.

Ниже приведено краткое сравнение по основным показателям:

Метод	Итераций	Вычислений f	Вычислений ∇f	Вычислений Гессиана
Ньютон + Армизо	7	28	28	7
BFGS	10	0'	22	0
Градиентный спуск (Армизо)	75	150	75	0
Нельдер–Мид (simplex)	130	650	—	—

В методе BFGS прямая оценка функции (f) в явном виде не используется при построении приближения матрицы, но линейный поиск по Армизо генерирует 2f- и 1Vf-оценку на итерацию.

Основные выводы:

Метод Ньютона + Армихо показал наименьшее число итераций, однако потребовал вычисления Гессиана на каждой итерации (16 оценок f и 4 оценок ∇f внутри hess_f), что делает его тяжёлым по вычислительным затратам при большой размерности.

BFGS уравнивал скорость сходимости и стоимость шага: понадобилось около 10 итераций и лишь оценки градиента, без прямого вычисления Гессиана.

Градиентный спуск с Армихо надёжно сходится, но требует существенно больше итераций и, соответственно, больше оценок функции и градиента.

Нельдер–Мид (нулевого порядка) потребовал наибольшего количества итераций и оценок f , однако не нуждается в градиентах, что полезно для негладких или шумных функций.

Таким образом, для задач малой и средней размерности наиболее эффективным оказался метод Ньютона (при условии невырожденного Гессиана), для более общих задач предпочтительнее BFGS, а нулевые методы стоит применять лишь при отсутствии доступных производных.